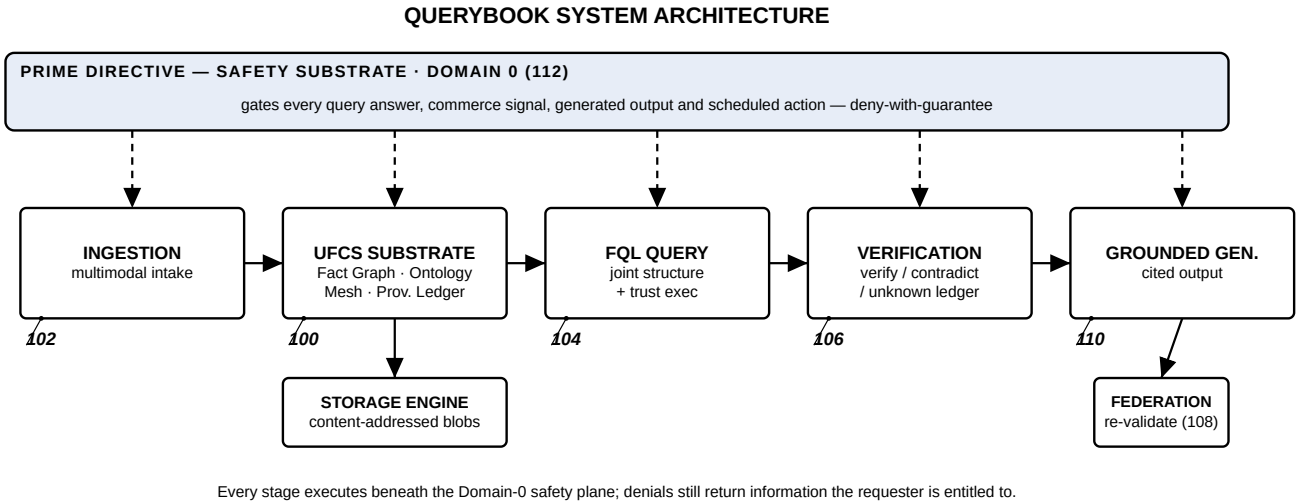
FIG. 1



Every stage executes beneath the Domain-0 safety plane; denials still return information the requester is entitled to.

FIG. 1

FIG. 1 — QueryBook system architecture (Prime Directive as Domain 0)

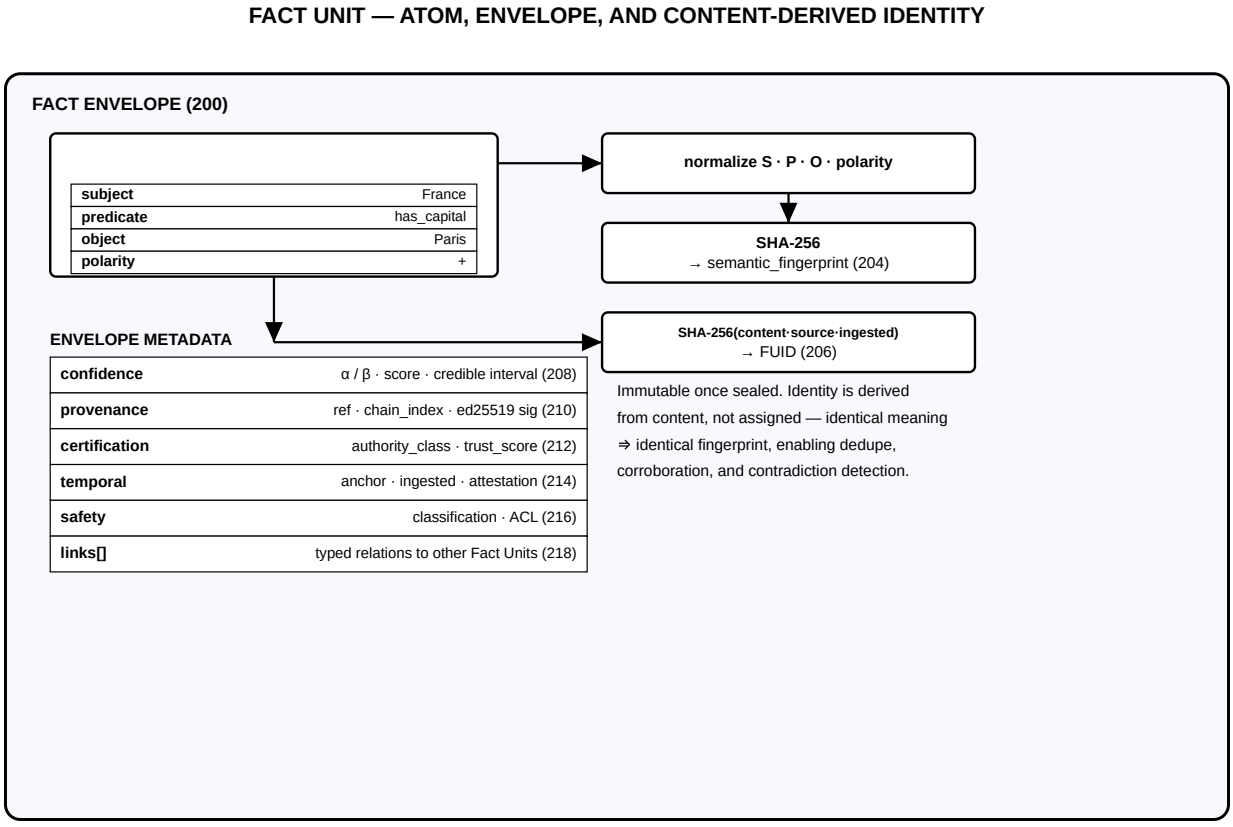
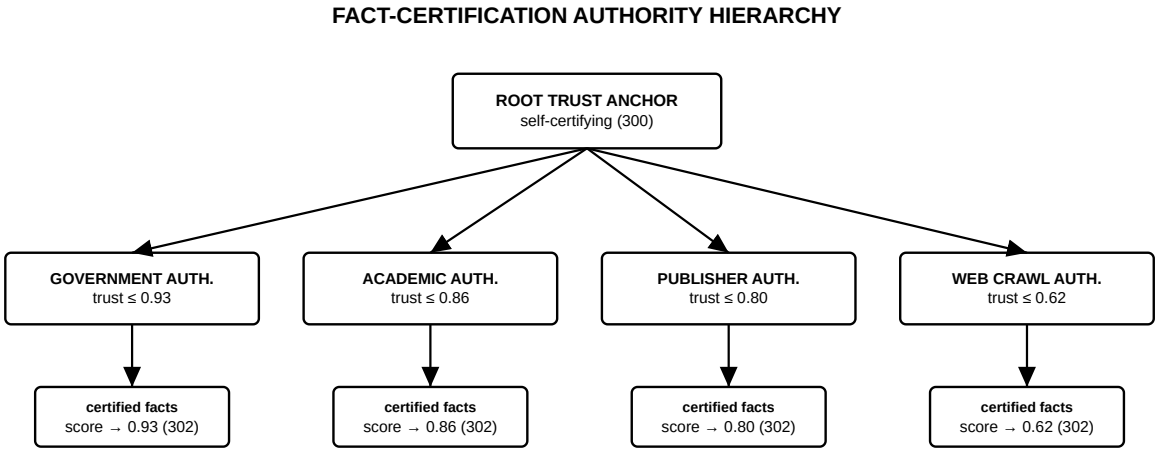


FIG. 2

FIG. 2 — Fact Unit — atom, envelope, and content-derived identity

FIG. 3



Trust flows downward; a fact inherits a ceiling from its issuing authority.

FIG. 3

FIG. 3 — Fact-certification authority hierarchy

FIG. 4

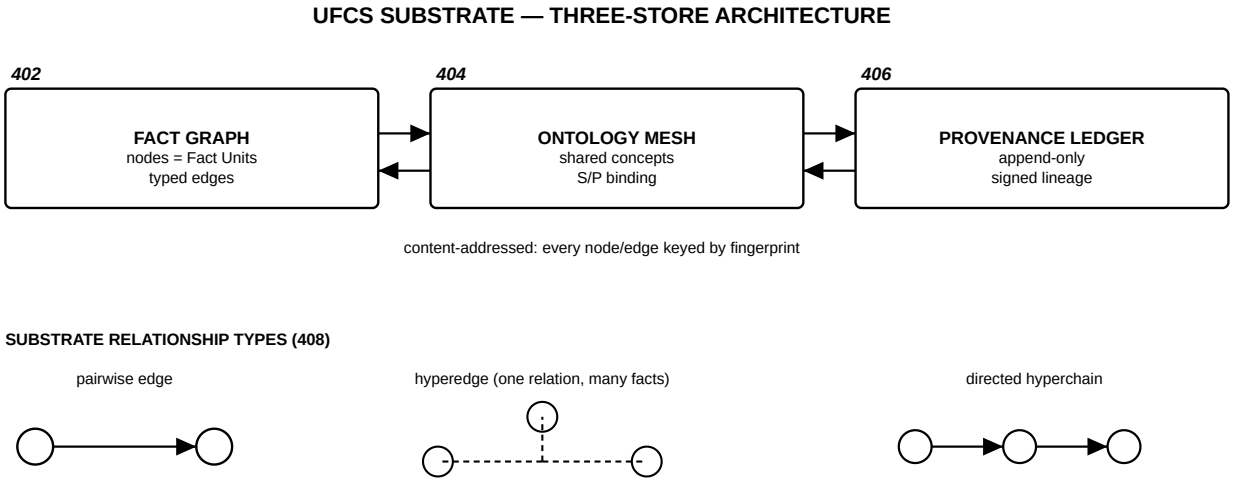


FIG. 4

FIG. 4 — UFCS substrate — three-store architecture

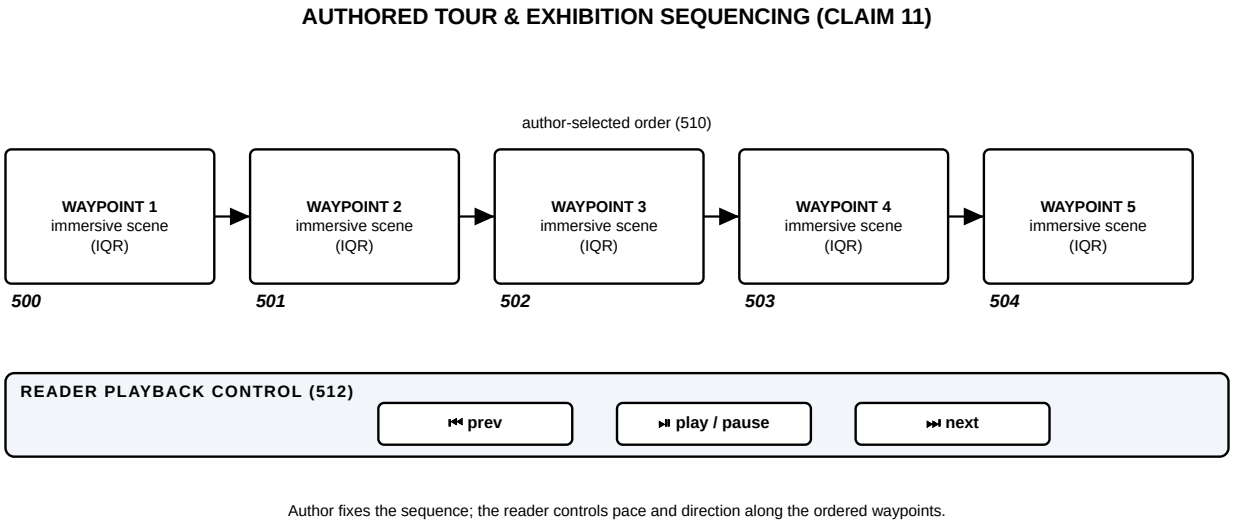
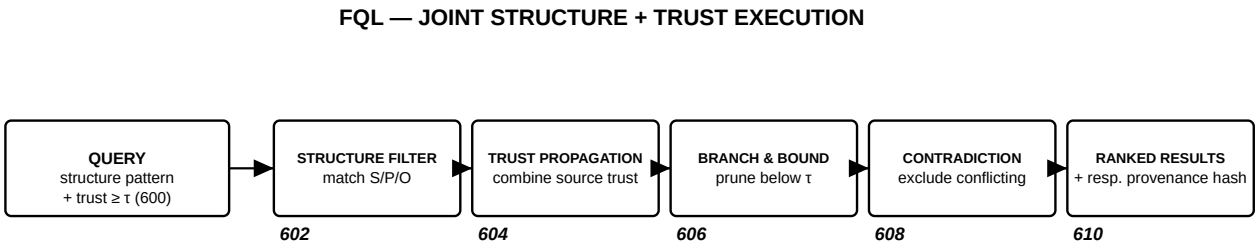


FIG. 5

FIG. 5 — Authored tour and exhibition sequencing

FIG. 6



Structure and certification are evaluated together, so pruning happens during traversal — not as a post-filter.

FIG. 6

FIG. 6 — FQL — joint structure and trust execution

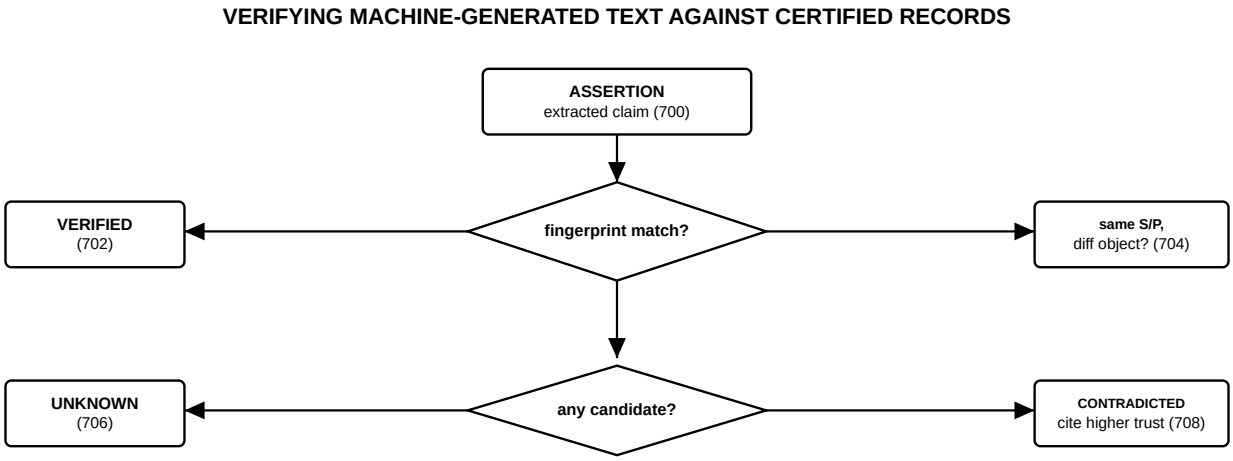
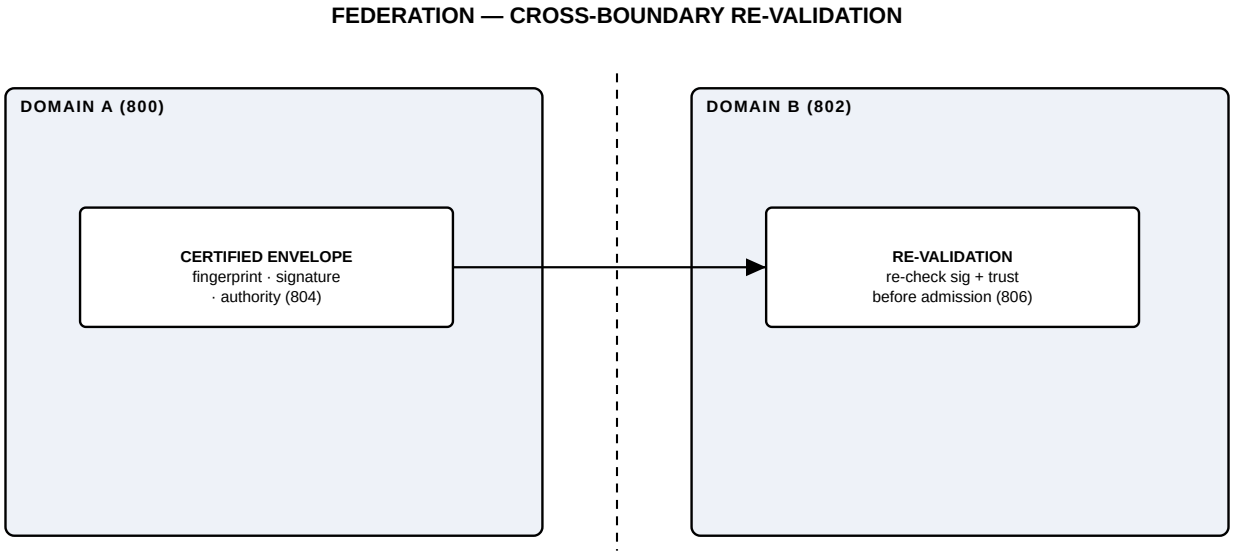


FIG. 7

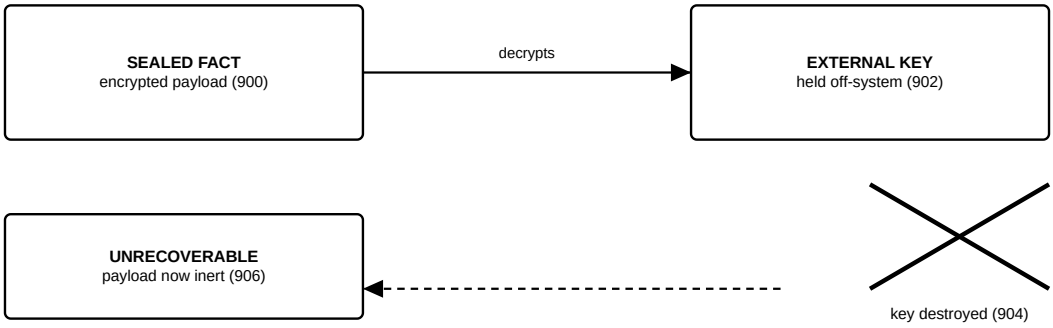
FIG. 7 — Verifying machine-generated text against certified records



A fact certified in one domain is re-validated by the receiver — never trusted blindly across the boundary.

FIG. 8

FACT-LEVEL ERASURE BY EXTERNAL-KEY DESTRUCTION



Erase one fact by destroying its key; the content-addressed store is otherwise append-only and immutable.

FIG. 9

FIG. 9 — Fact-level erasure by external-key destruction

FIG. 10

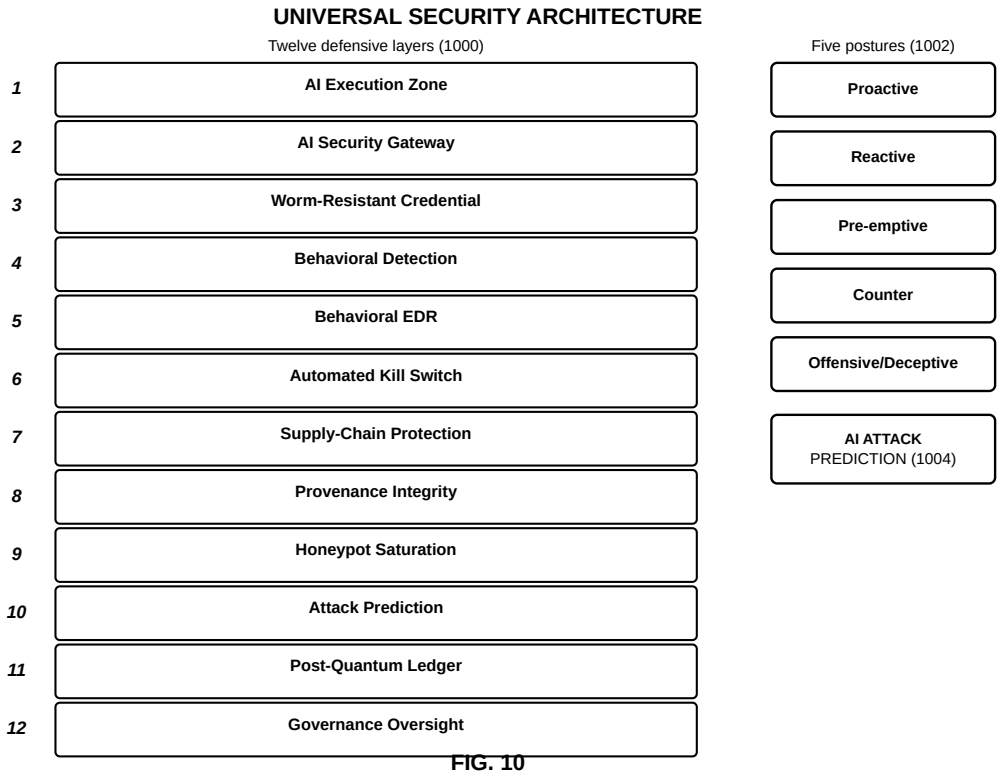


FIG. 10

FIG. 10 — Universal security architecture

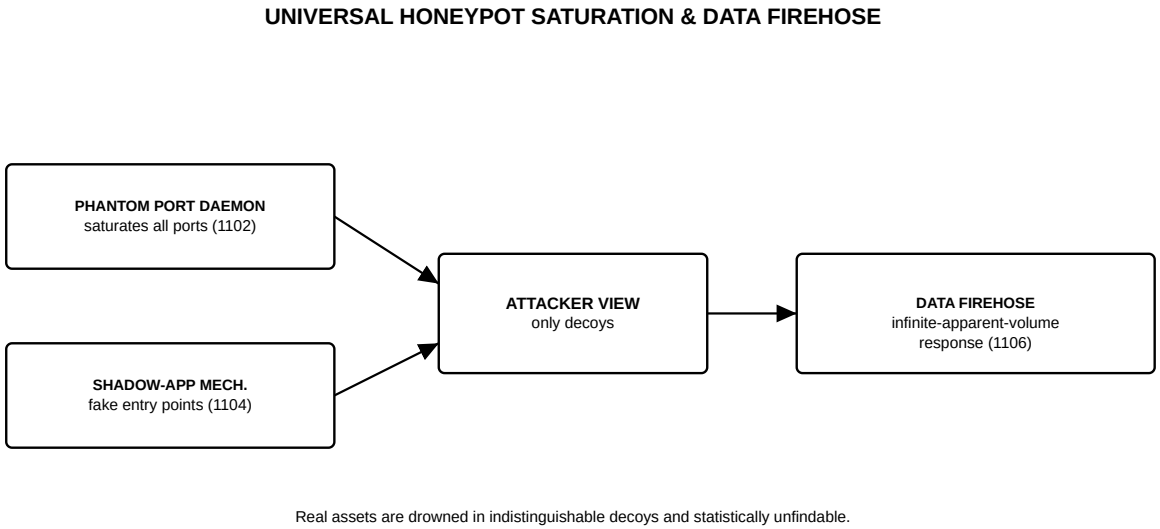
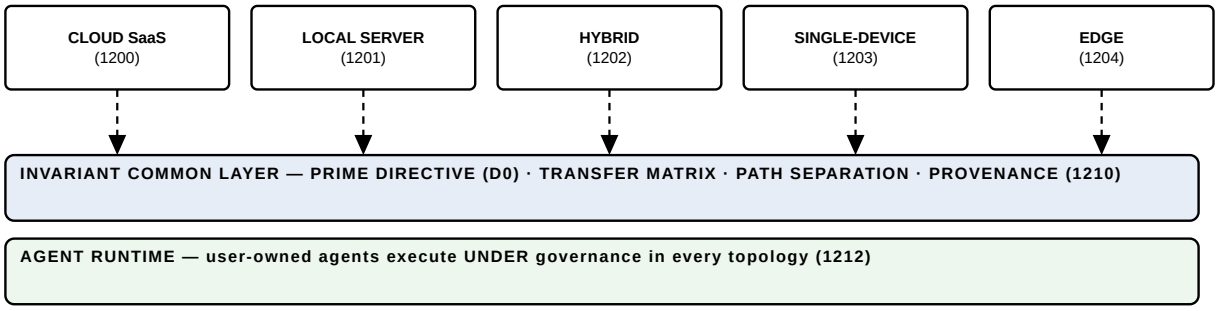


FIG. 11

FIG. 11 — Universal honeypot saturation and data firehose

DEPLOYMENT TOPOLOGIES OVER AN INVARIANT COMMON LAYER



The topology varies; the safety/governance/provenance layer and the governed agent runtime are identical across all five.

FIG. 12

FIG. 12 — Deployment topologies over an invariant common layer

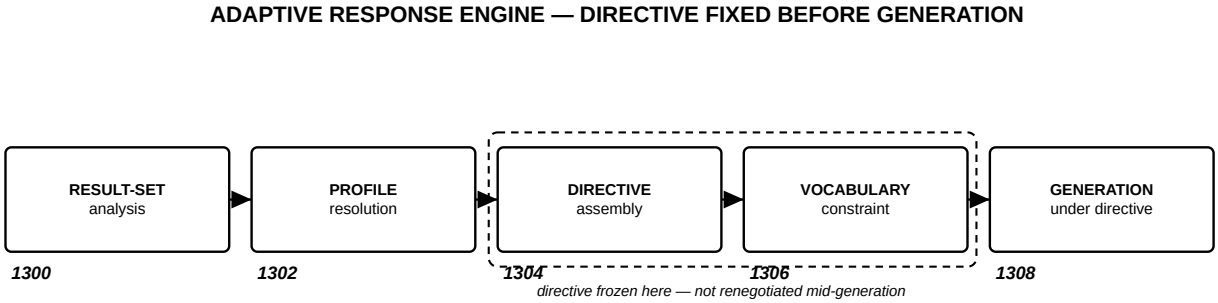
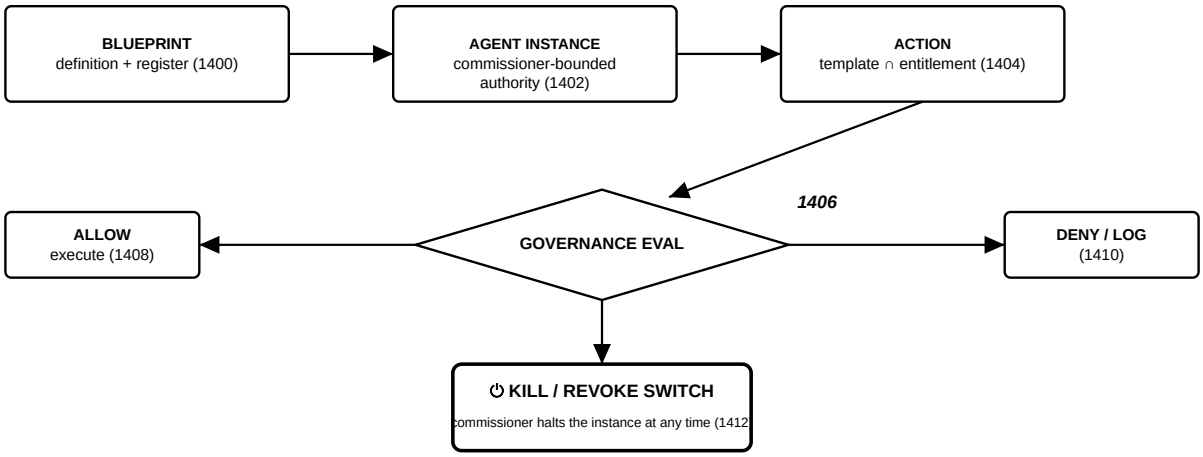

ADAPTIVE RESPONSE ENGINE — DIRECTIVE FIXED BEFORE GENERATION
RESULT-SET
analysis
PROFILE
resolution
DIRECTIVE
assembly
VOCABULARY
constraint
GENERATION
under directive
1300
1302
1304
1306
1308
directive frozen here — not renegotiated mid-generation

FIG. 13

USER-OWNED AGENT SYSTEM WITH GOVERNANCE & KILL SWITCH

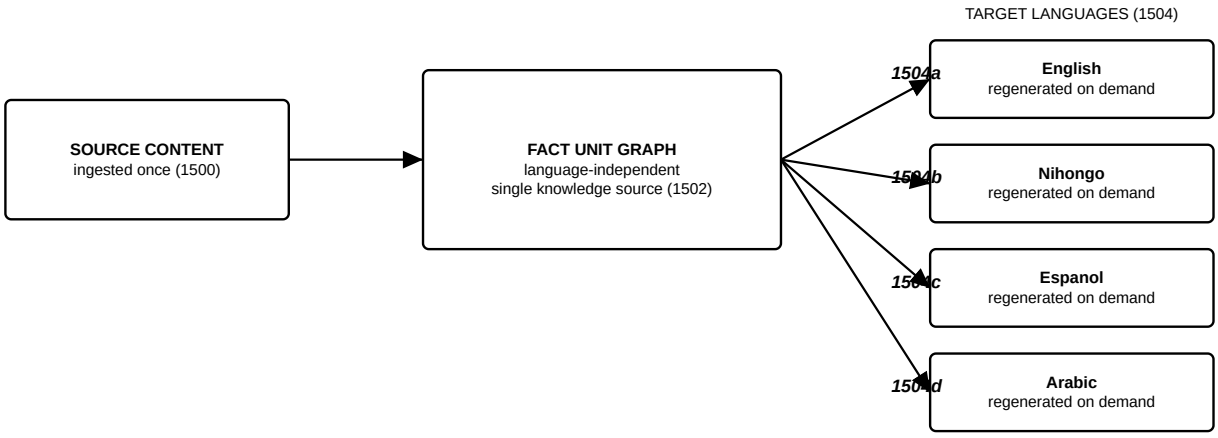


Governance is checked at every action; an agent can never exceed its commissioner, who holds an unconditional kill control.

FIG. 14

FIG. 14 — User-owned agent system with governance and kill switch

DETERMINISTIC MULTI-LANGUAGE REGENERATION

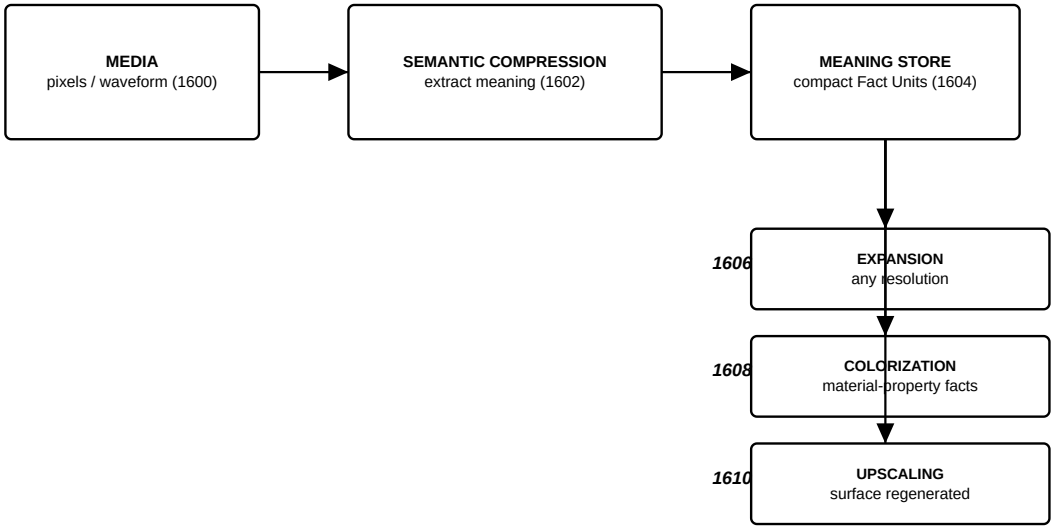


One graph, many surfaces: no per-language re-ingestion; each language is a projection of the same facts.

FIG. 15

FIG. 15 — Deterministic multi-language regeneration

SEMANTIC COMPRESSION, EXPANSION, COLORIZATION & UPSCALING

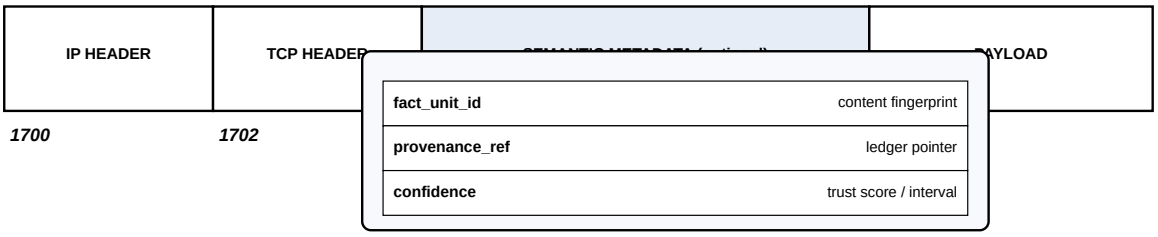


Store meaning, not surface; compression and regeneration are the same primitive — no generational quality loss.

FIG. 16

FIG. 16 — Semantic compression, expansion, colorization and upscaling

QUERYBOOK-ENHANCED TCP/IP HYBRID PROTOCOL PACKET



A non-participating endpoint ignores the semantic header — provenance travels with the bits, backward-compatible.

FIG. 17

FIG. 17 — QueryBook-enhanced TCP/IP hybrid protocol packet

NEUROMORPHIC COGNITIVE LAYER — ATTRACTOR CONVERGENCE

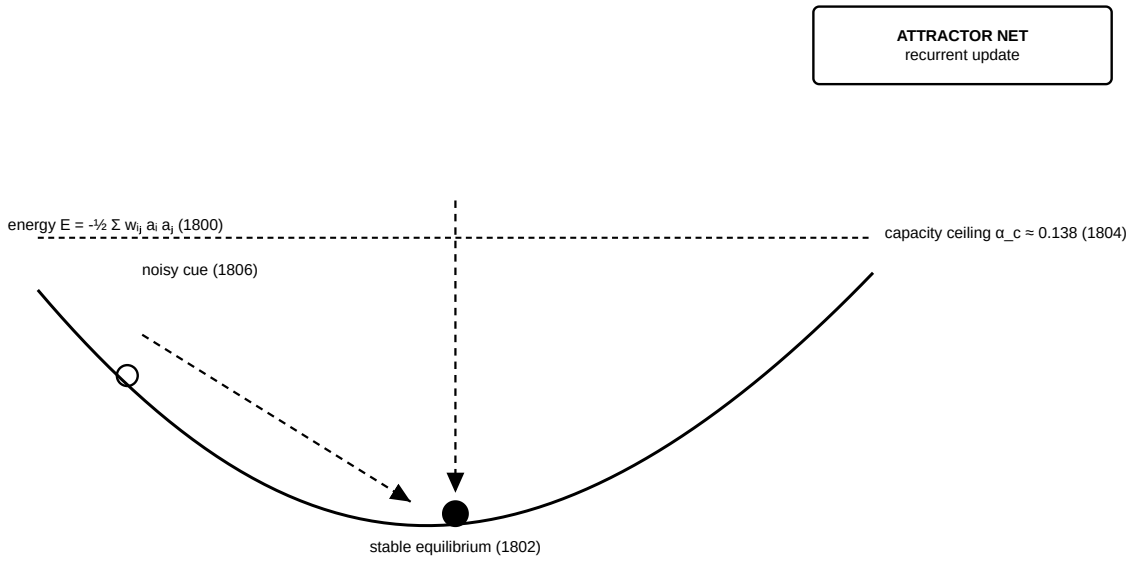
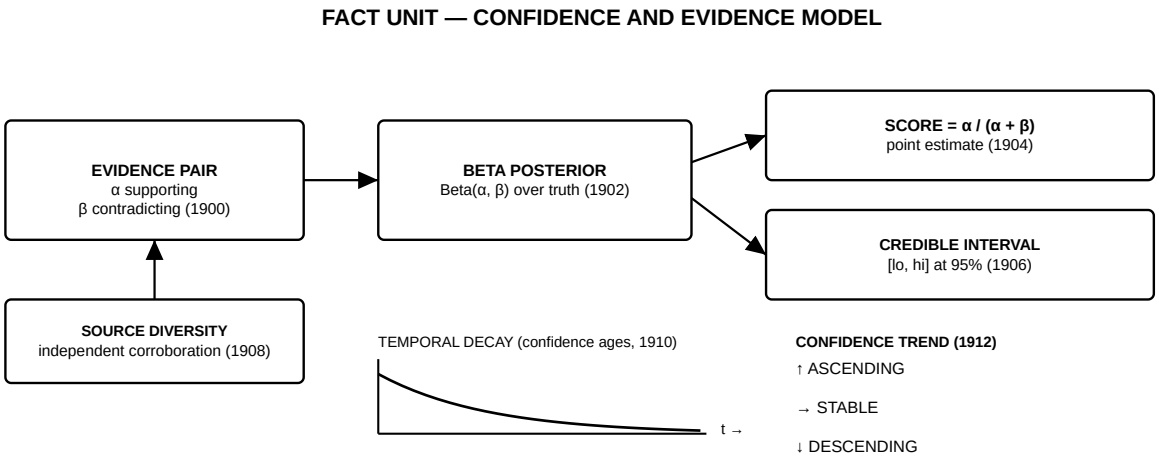


FIG. 18

A partial/noisy cue relaxes down the energy surface to the nearest stored equilibrium — recall as convergence, bounded by α_c.

FIG. 18 — Neuromorphic cognitive layer — attractor convergence



Trust is a governed evidence estimate — not a single number: it carries an interval, ages over time, and reports its direction.

FIG. 19

FIG. 19 — Fact Unit — confidence and evidence model

FIG. 20

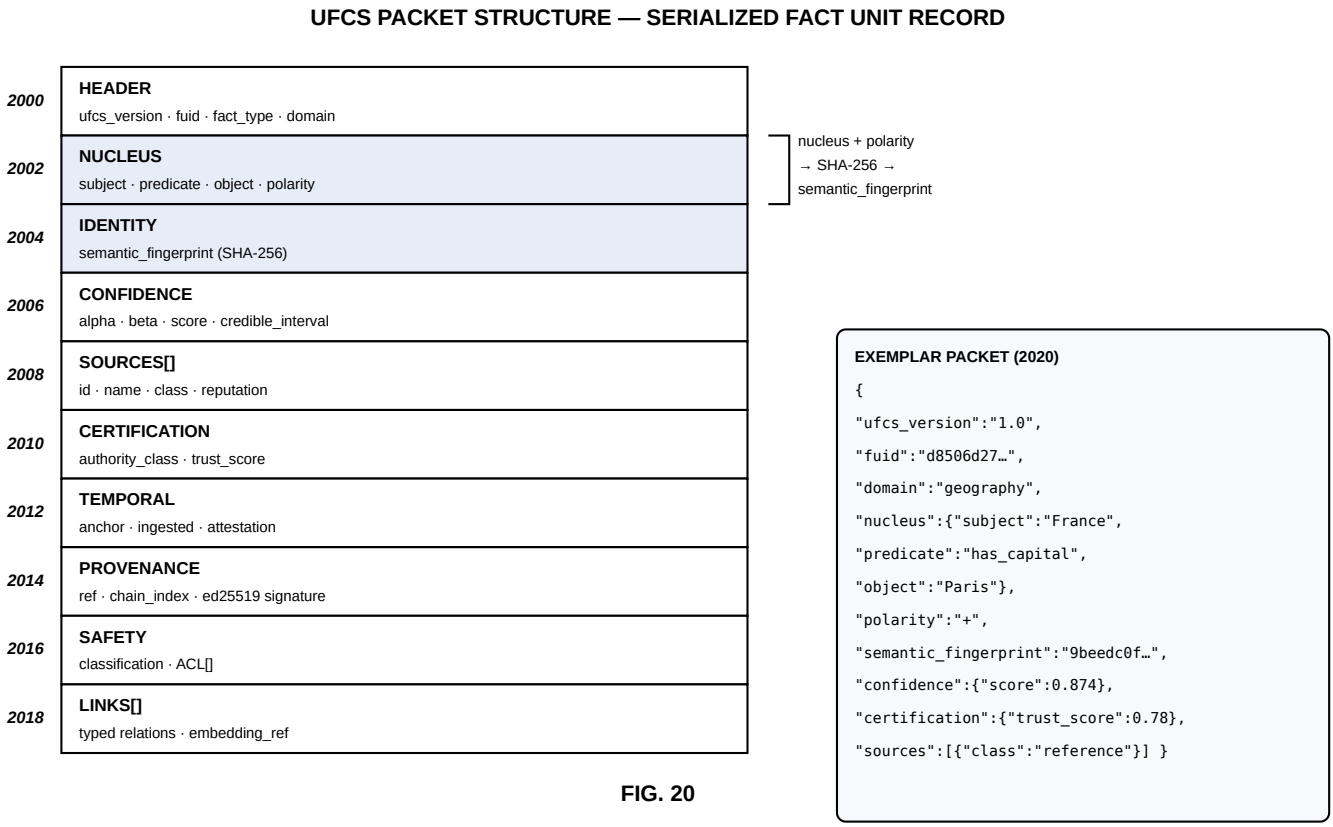
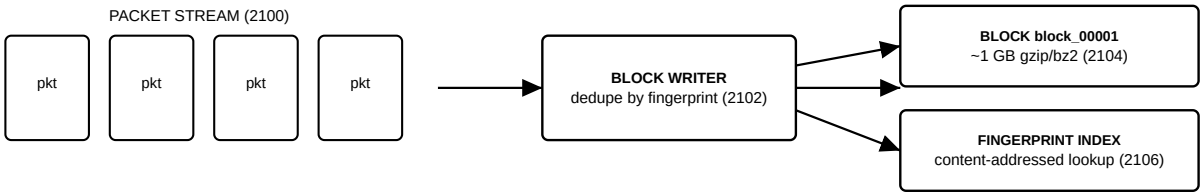


FIG. 20

FIG. 20 — UFCS packet structure — serialized Fact Unit record

UFCS PACKET FRAMING — CONTENT-ADDRESSED BLOCK STORAGE



Immutable packets are appended to ~1 GB content-addressed blocks; a fingerprint index gives O(1) dedupe and retrieval.

Identical meaning ⇒ identical fingerprint ⇒ stored once; corroboration and contradiction are detected at ingest.

FIG. 21

FIG. 22

FQL — QUERY ANATOMY AND WORKED EXAMPLE

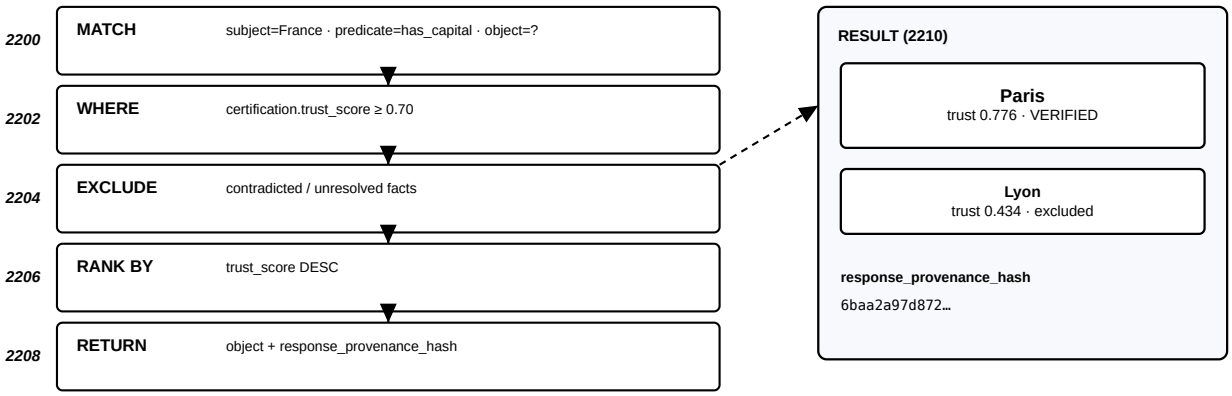


FIG. 22

Structure and trust are one query; the engine returns the highest-trust answer with a hash binding the result to its evidence.

FIG. 22 — FQL — query anatomy and worked example

FIG. 23

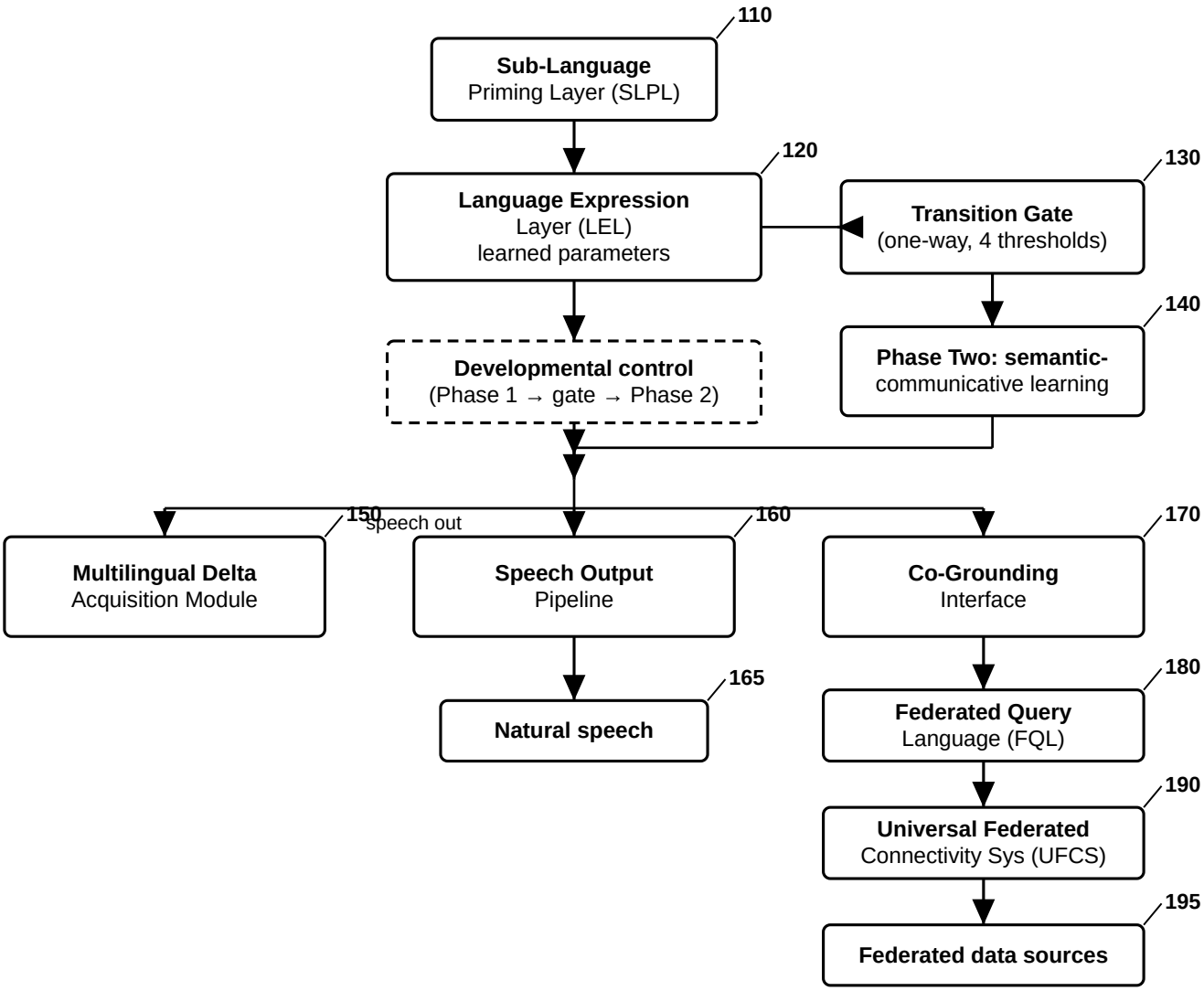


FIG. 23 — QueryBook LEL — Overall System Architecture

FIG. 24

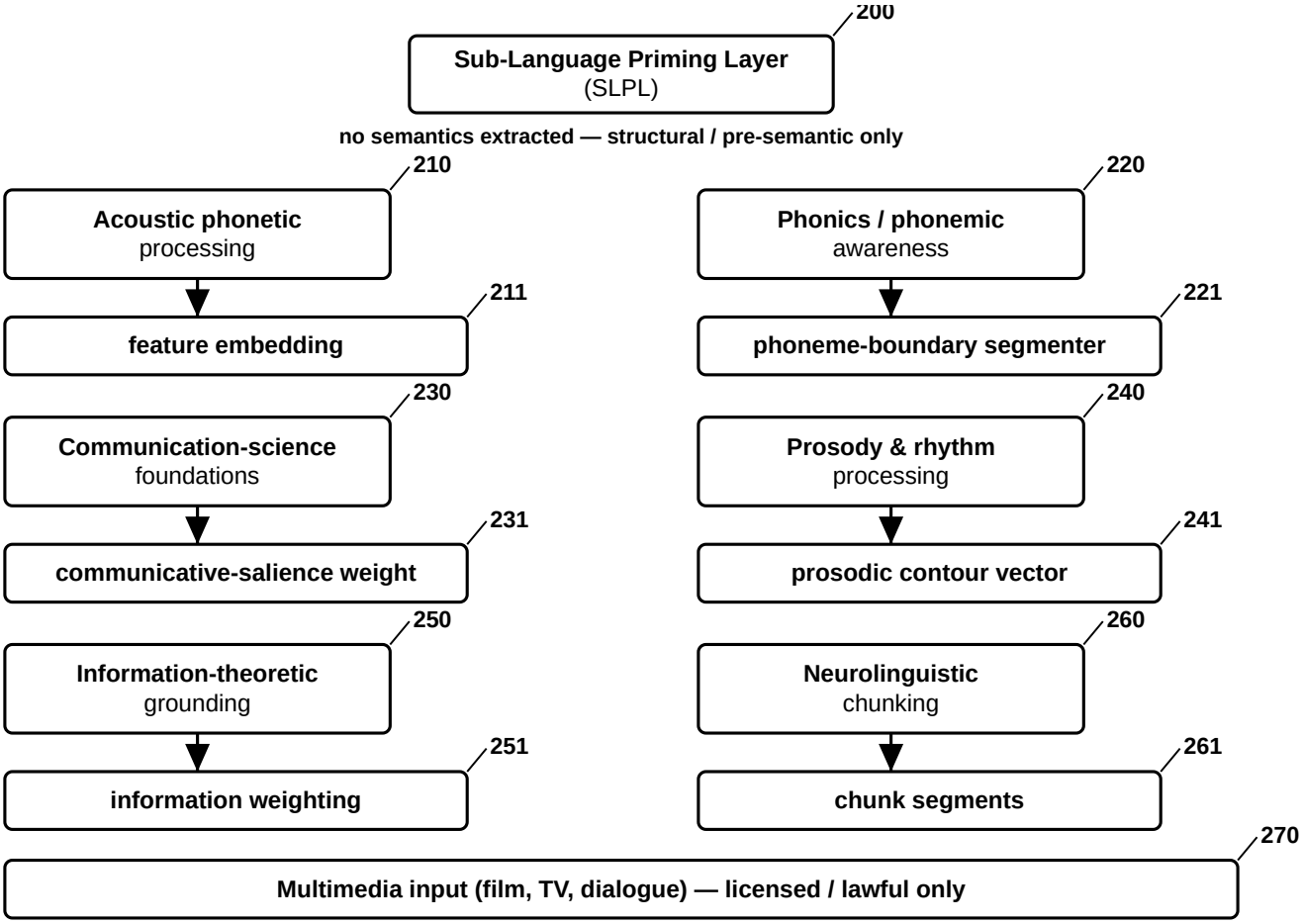


FIG. 24 — Sub-Language Priming Layer (SLPL) and Its Measurable Outputs

FIG. 25

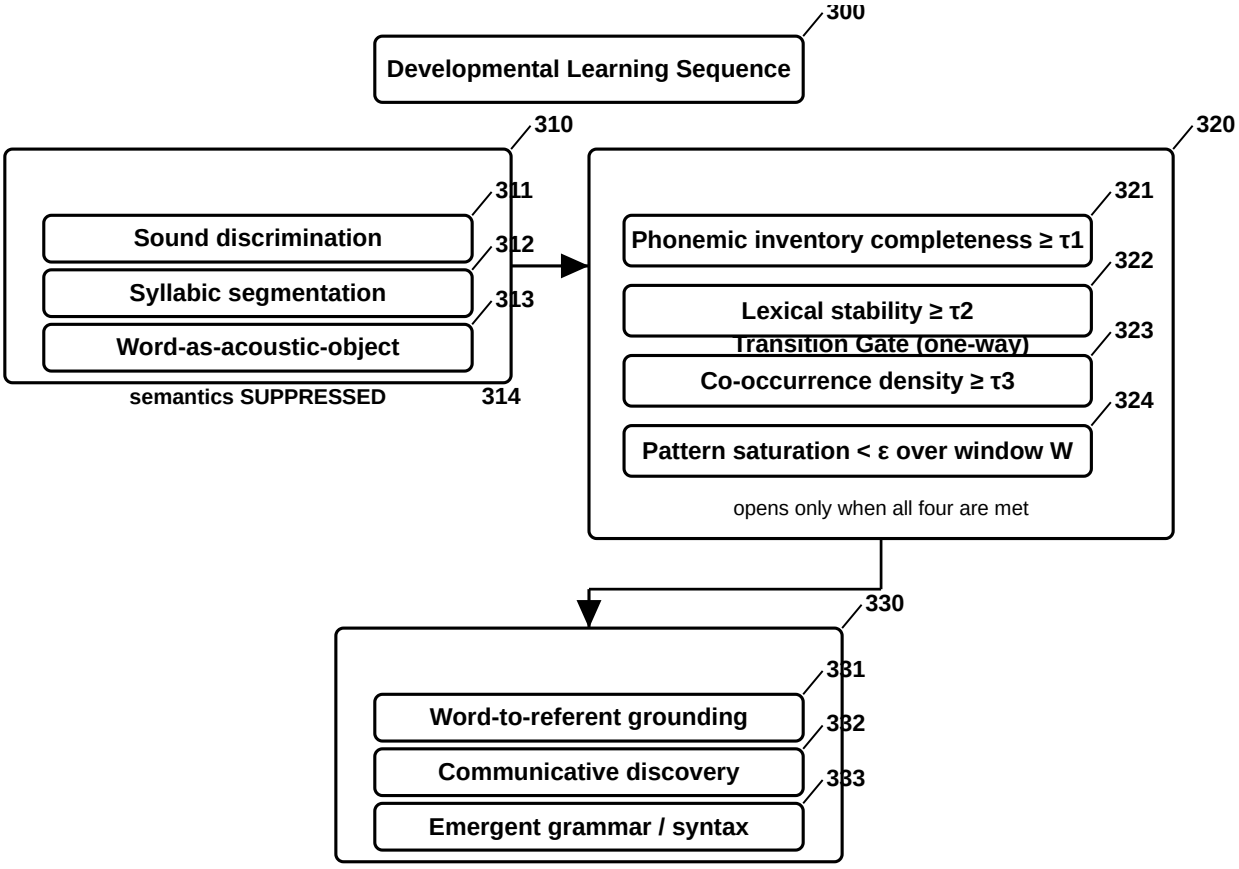


FIG. 25 — Developmental Learning Sequence and One-Way Transition Gate

FIG. 26

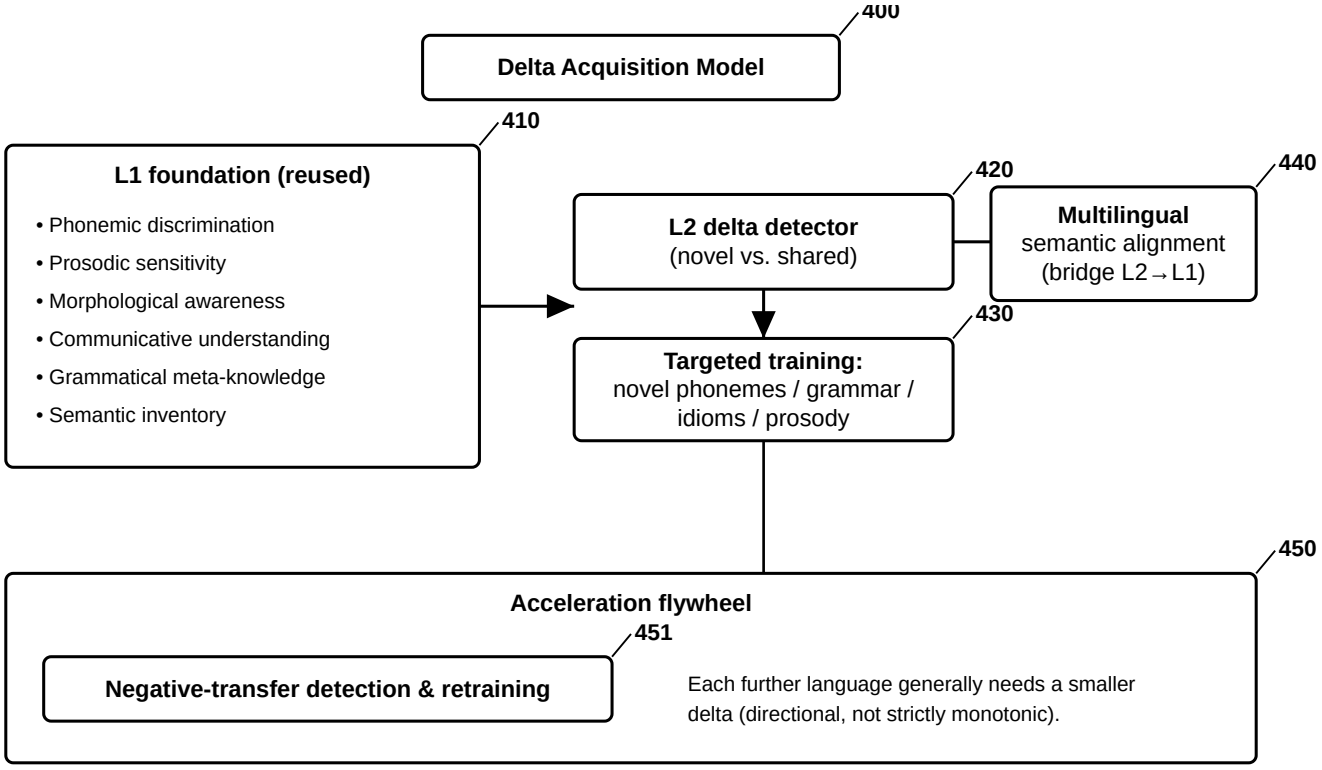


FIG. 26 — Multilingual Transfer — Delta Acquisition and Acceleration Flywheel

FIG. 27

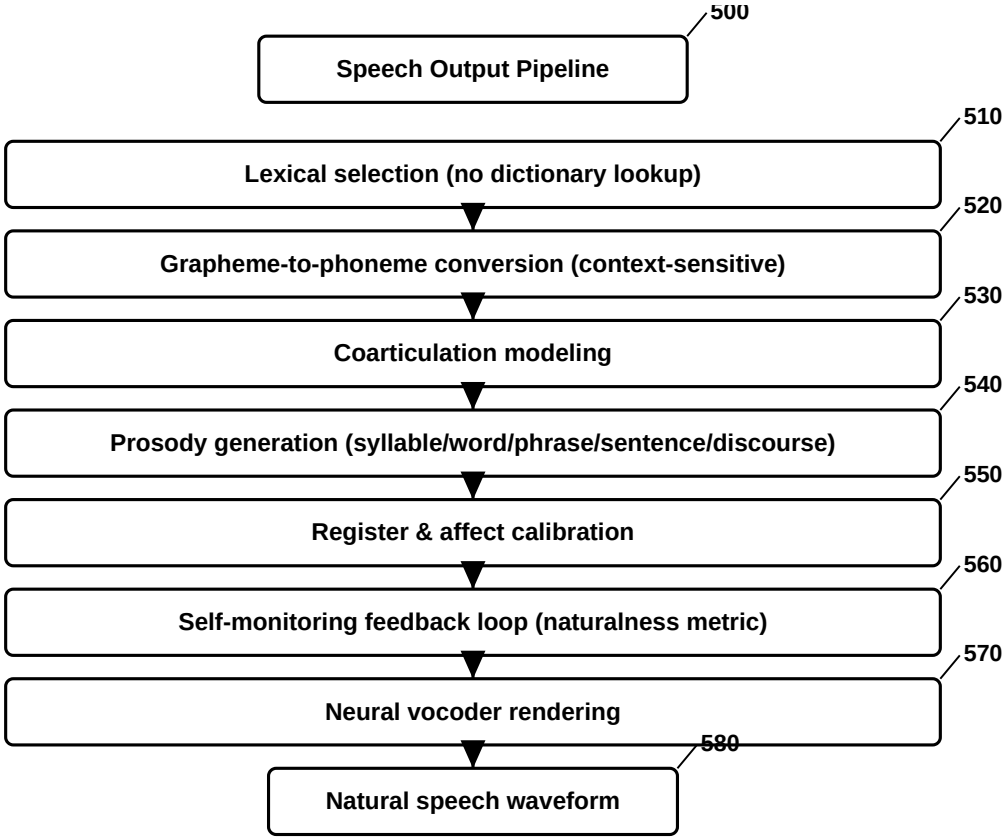


FIG. 27 — Speech Output Pipeline (Seven Learned Stages)

FIG. 28

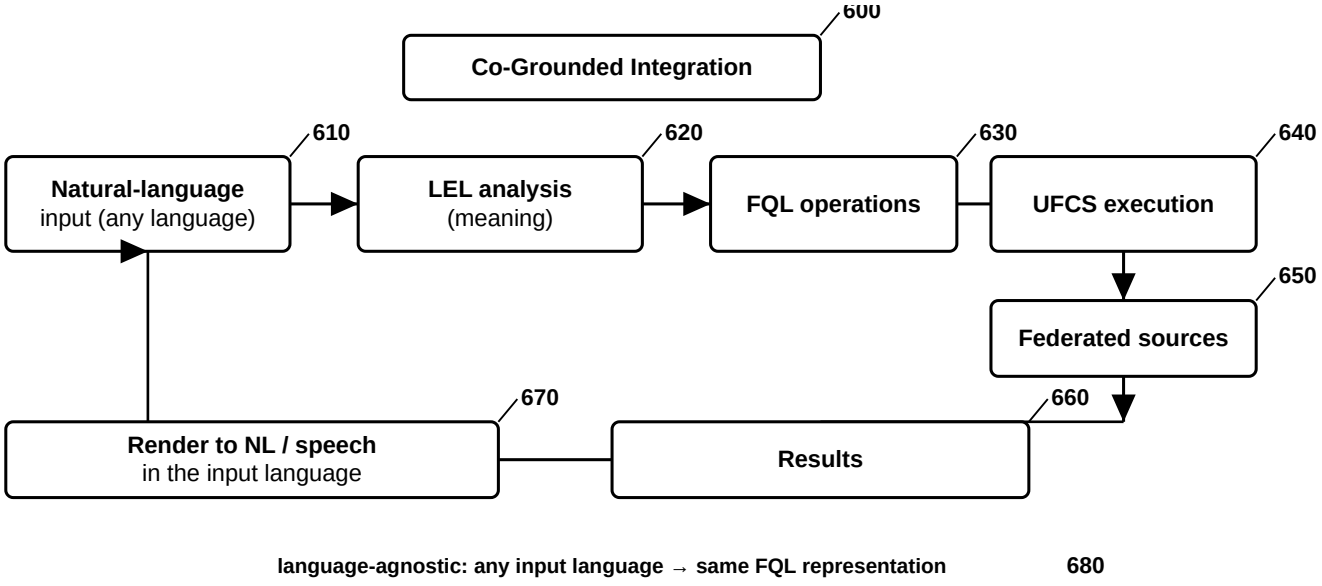


FIG. 28 — Co-Grounded Integration with FQL and UFCS

FIG. 29

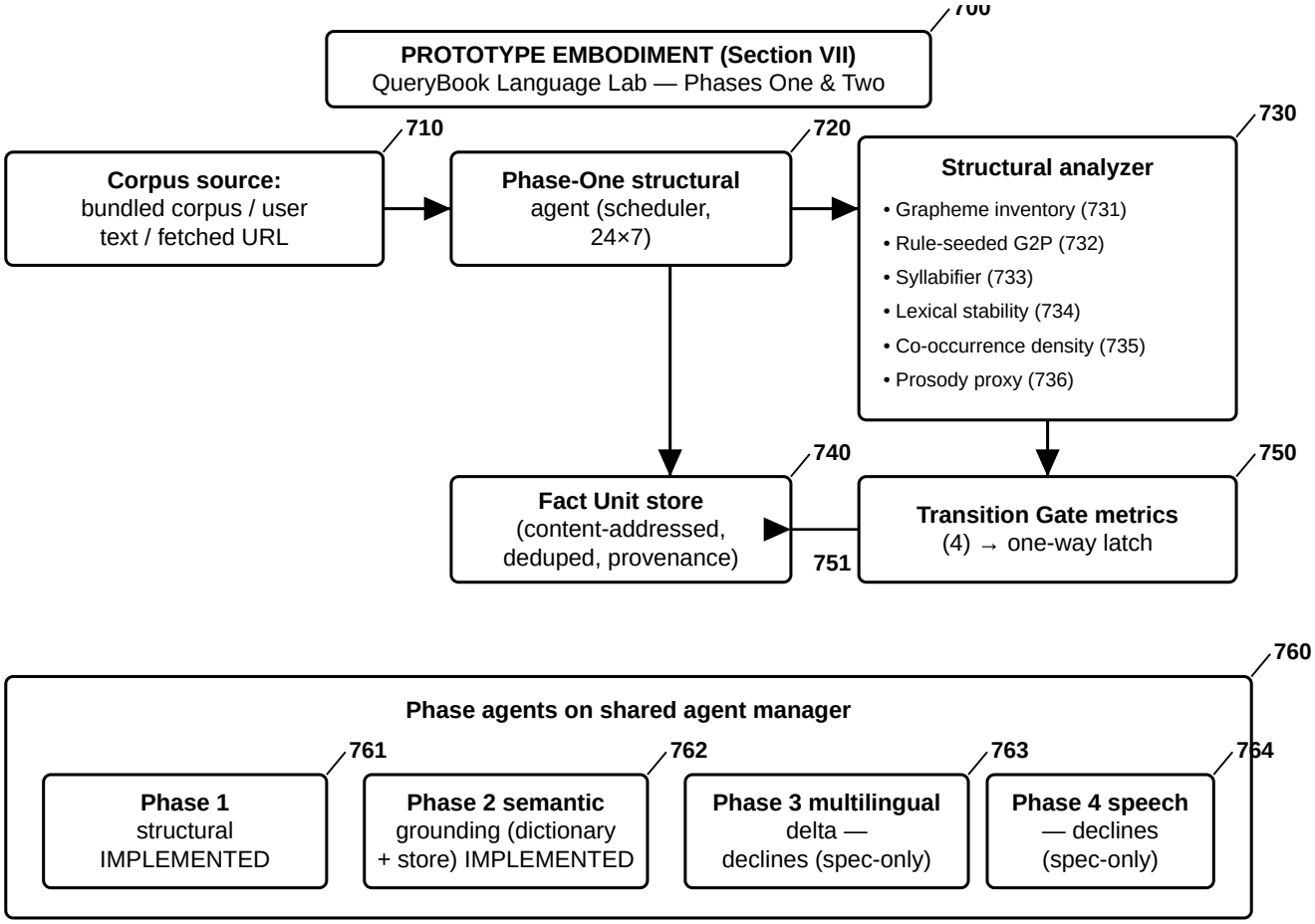


FIG. 29 — Reduced-to-Practice Embodiment — Language Lab Phases One and Two Prototype

FIG. 30

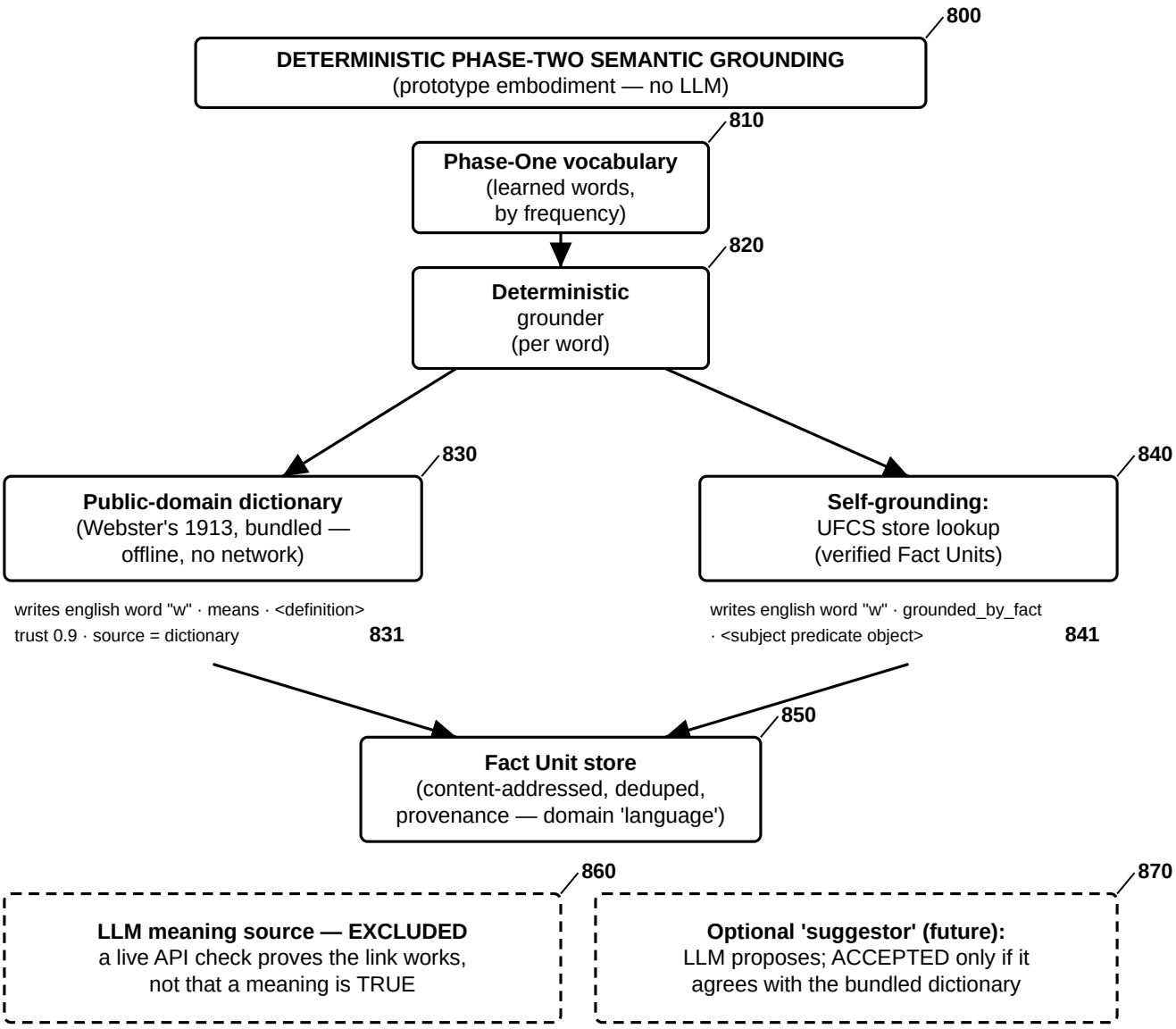


FIG. 30 — Deterministic Phase-Two Semantic Grounding — Dictionary and Store (No LLM)

FIG. 31

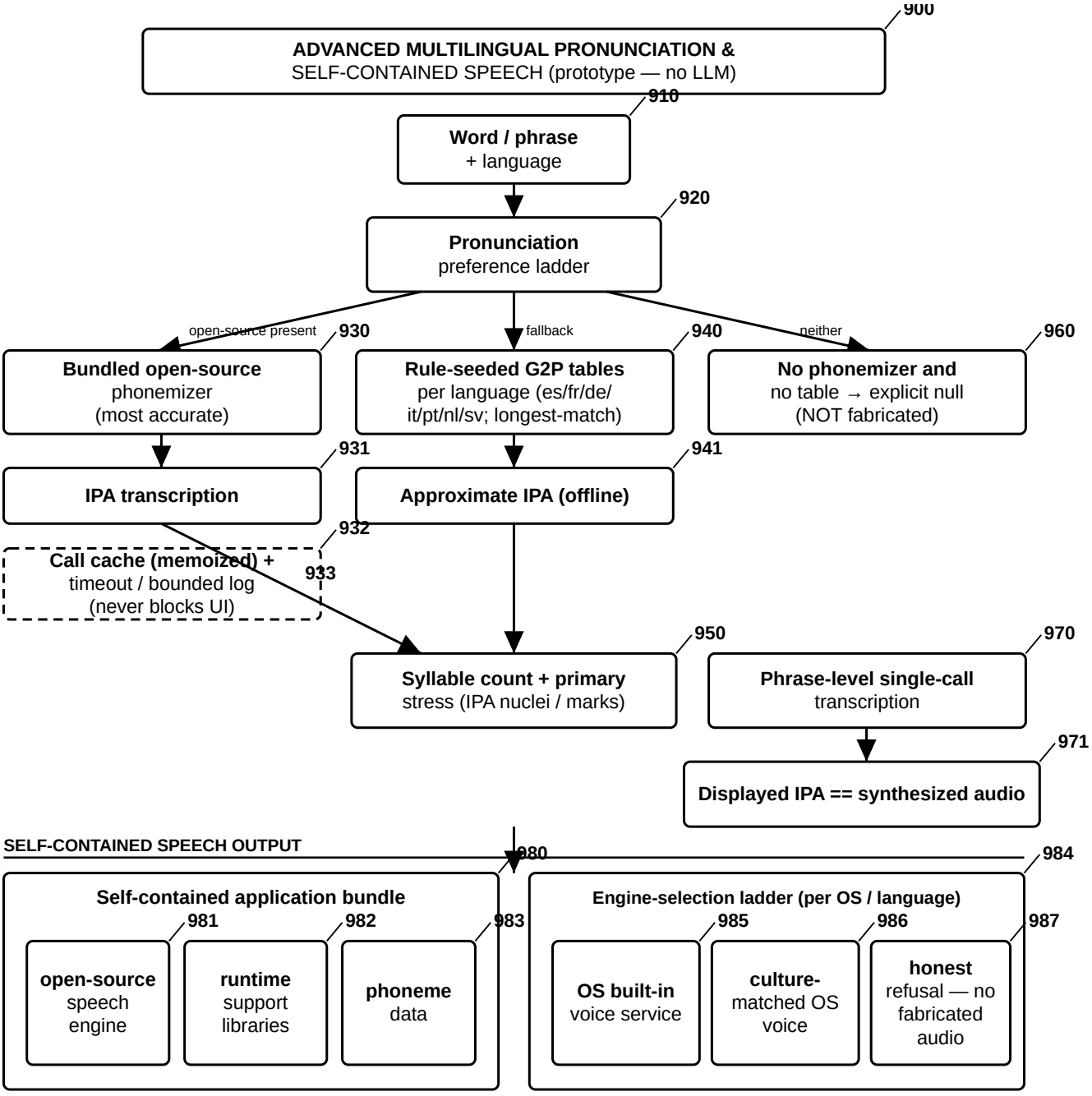


FIG. 31 — Advanced Multilingual Pronunciation and Self-Contained Speech (No LLM)

FIG. 32

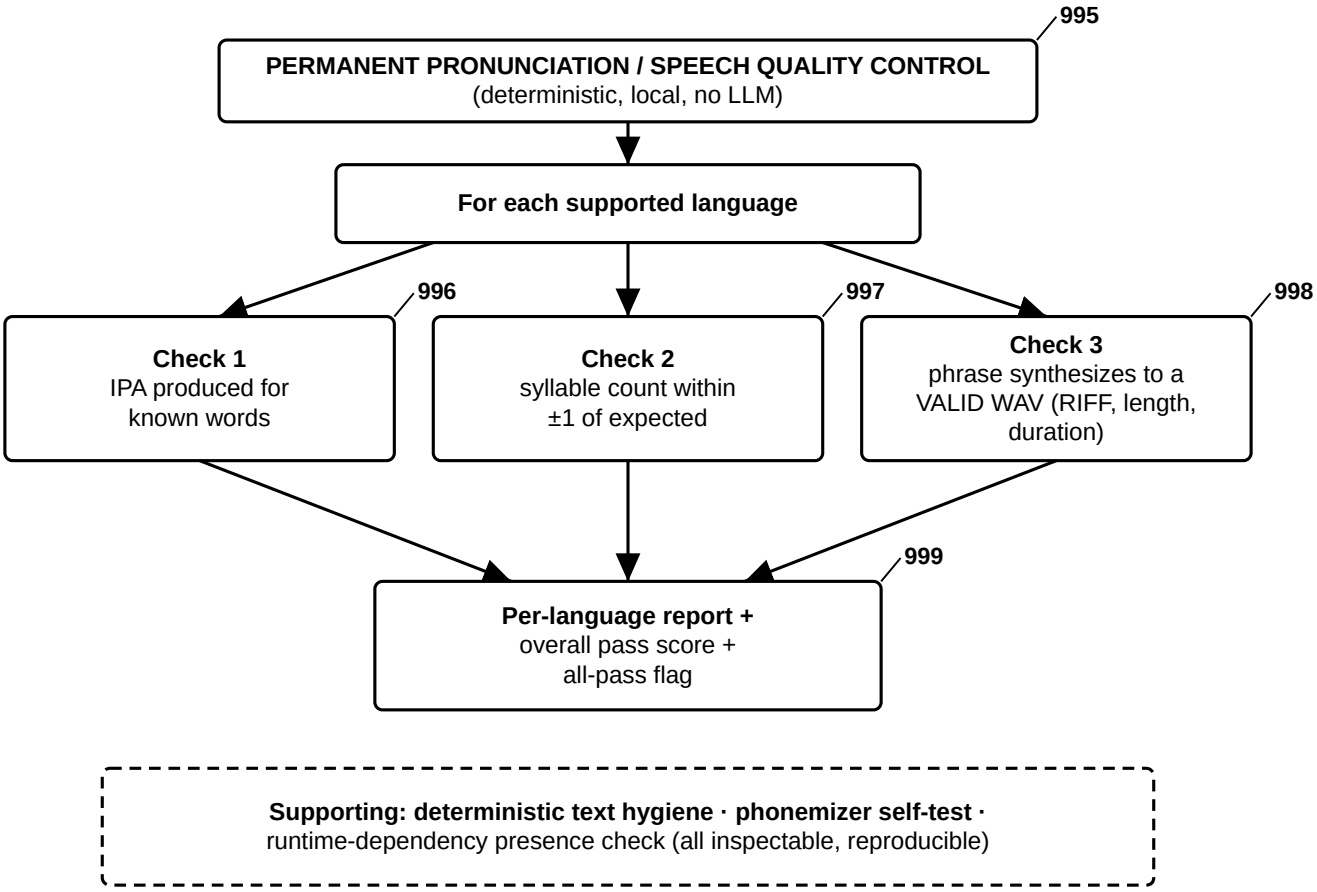


FIG. 32 — Permanent Pronunciation and Speech Quality-Control Battery (No LLM)

FIG. 33

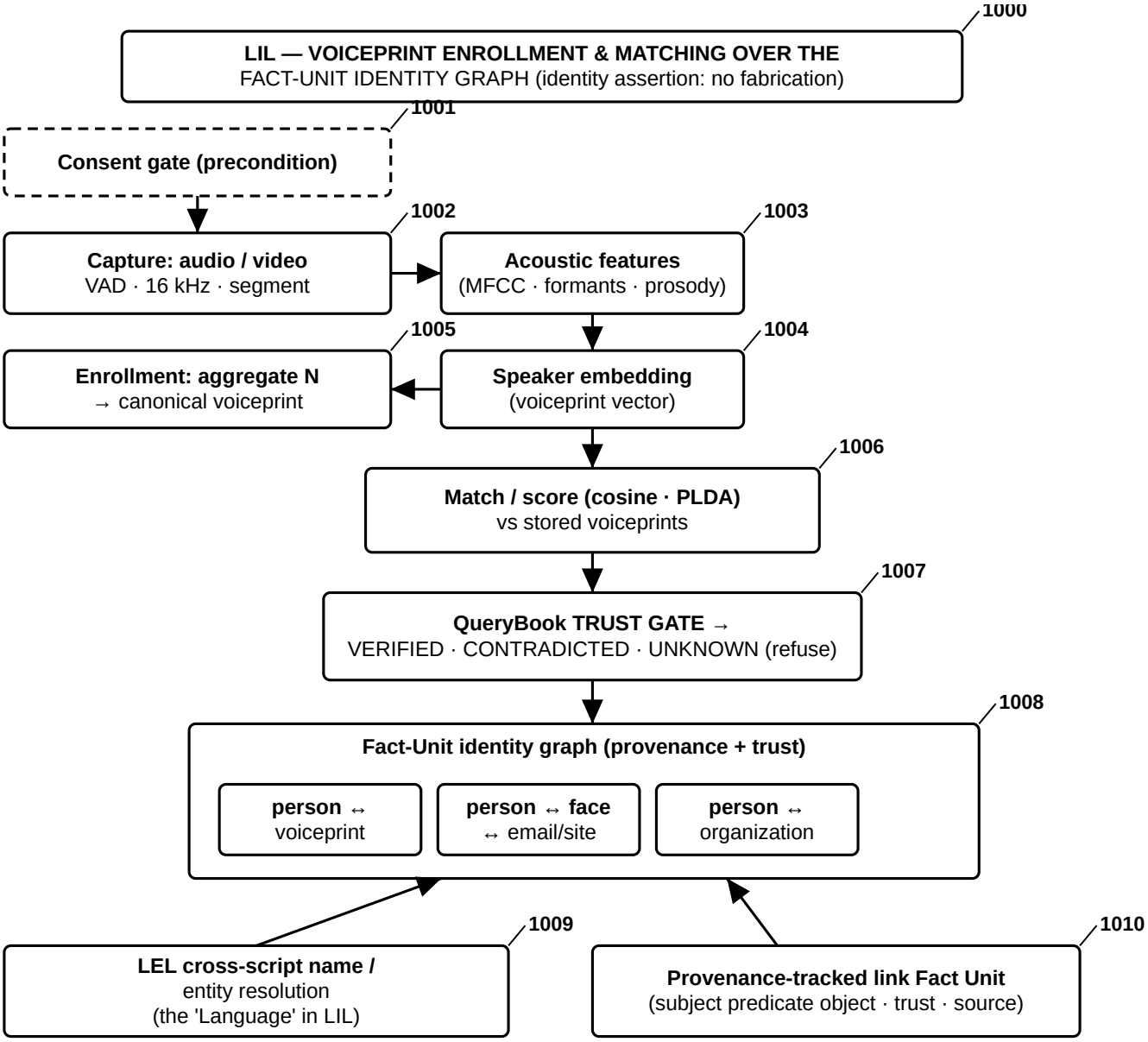


FIG. 33 — LIL — Voiceprint Enrollment and Matching over the Fact-Unit Identity Graph

FIG. 34

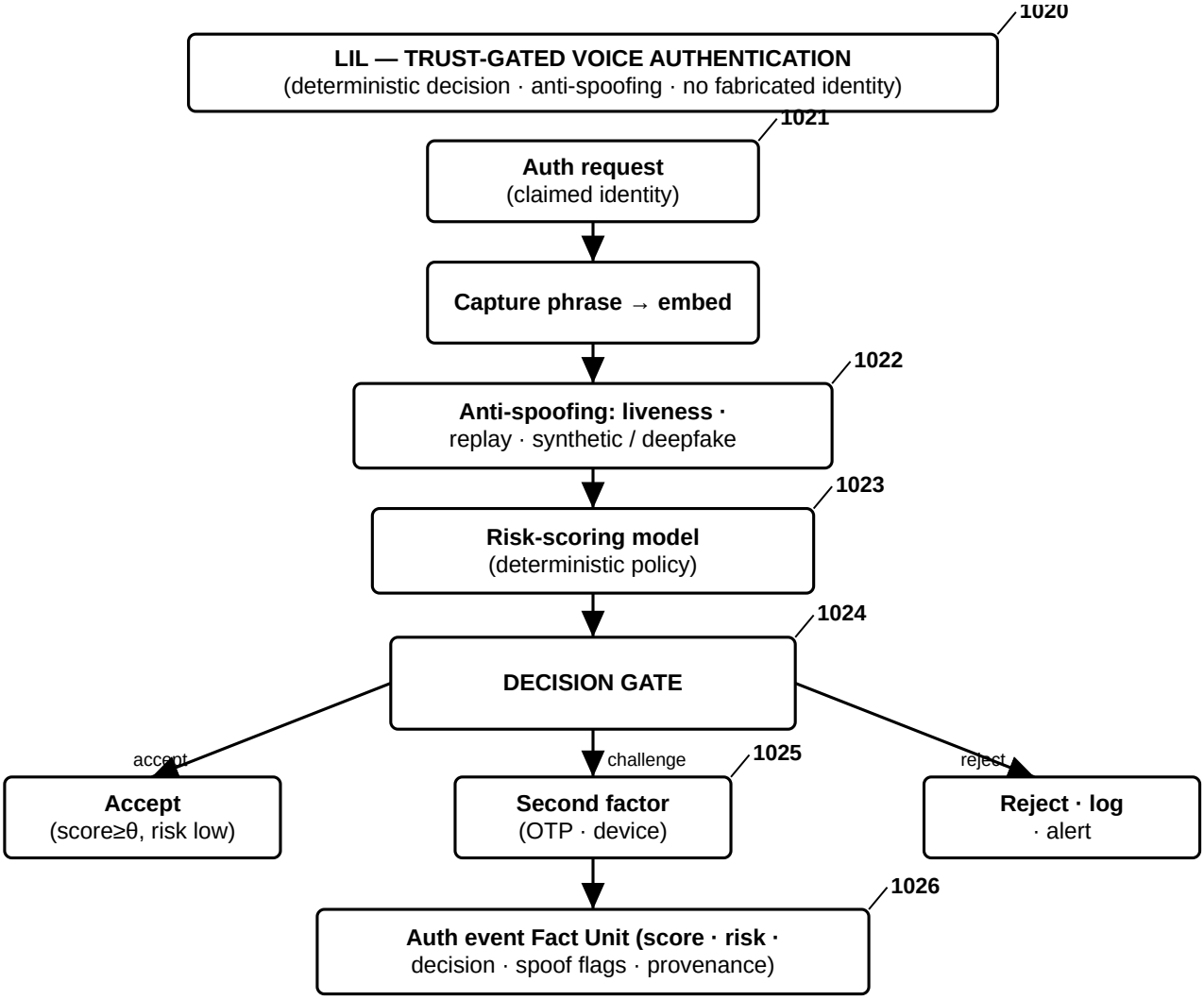


FIG. 34 — LIL — Trust-Gated Voice Authentication with Anti-Spoofing

FIG. 35

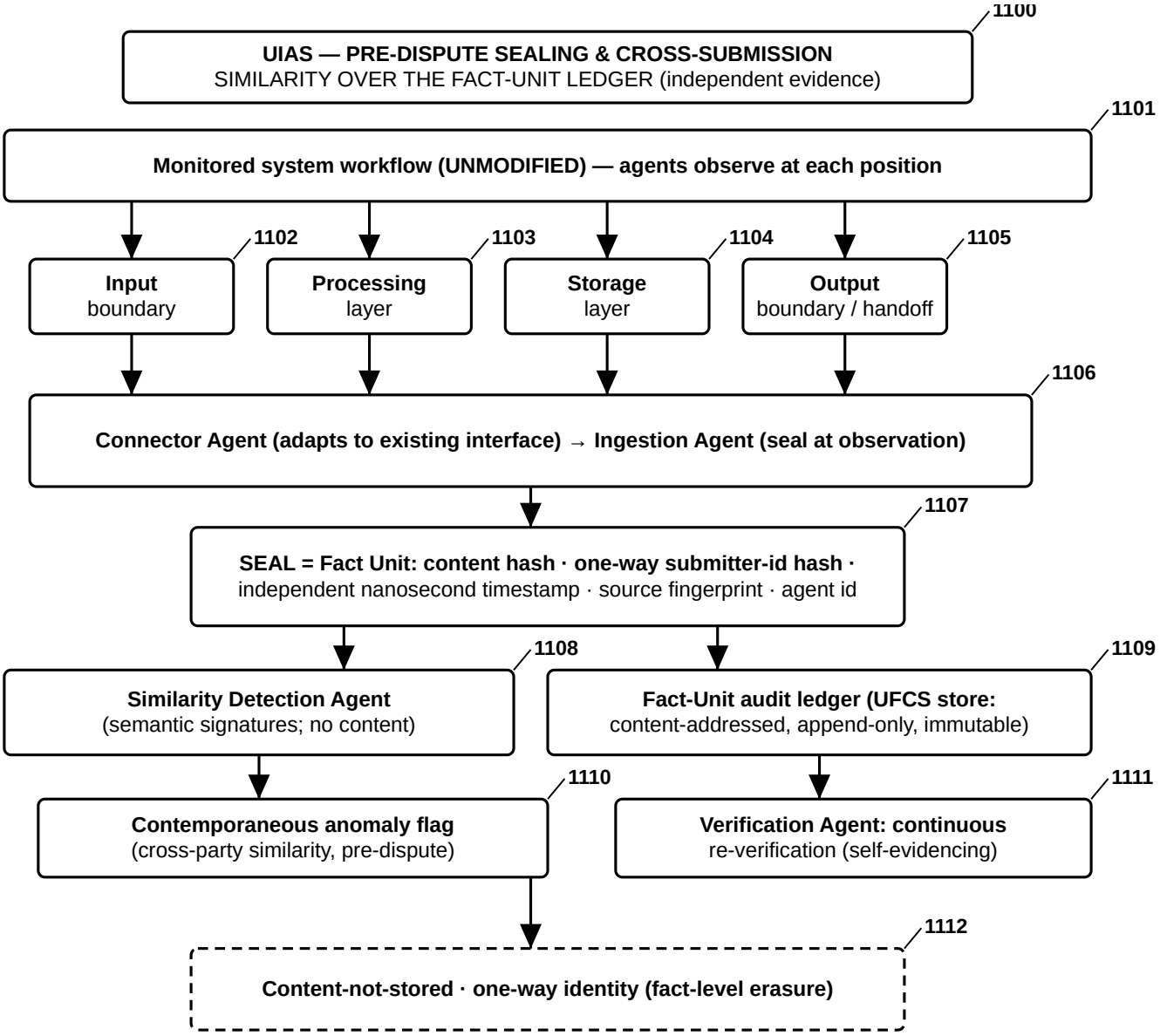


FIG. 35 — UIAS — Pre-Dispute Sealing and Cross-Submission Similarity over the Fact-Unit Ledger

FIG. 36

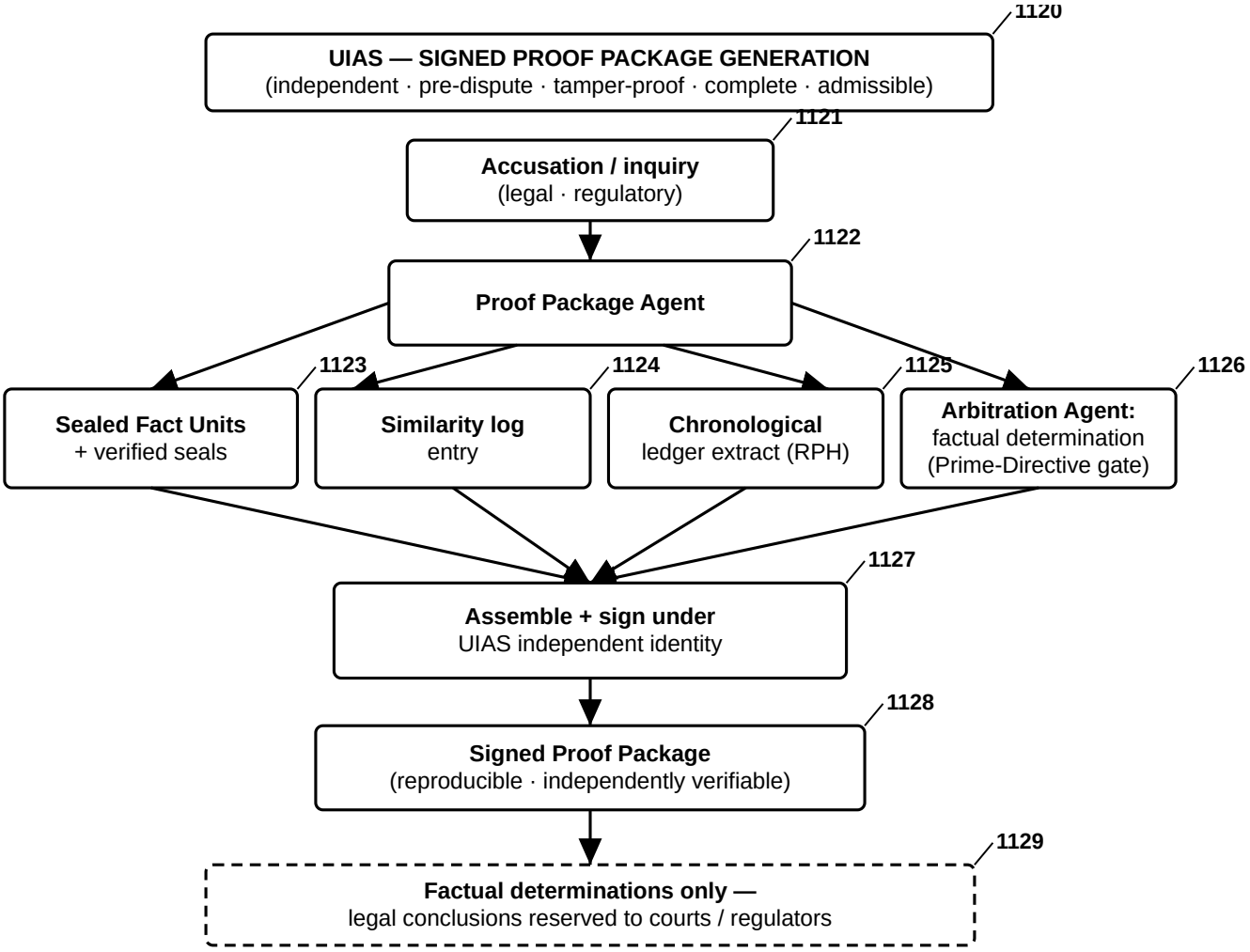


FIG. 36 — UIAS — Signed Proof Package Generation

FIG. 37

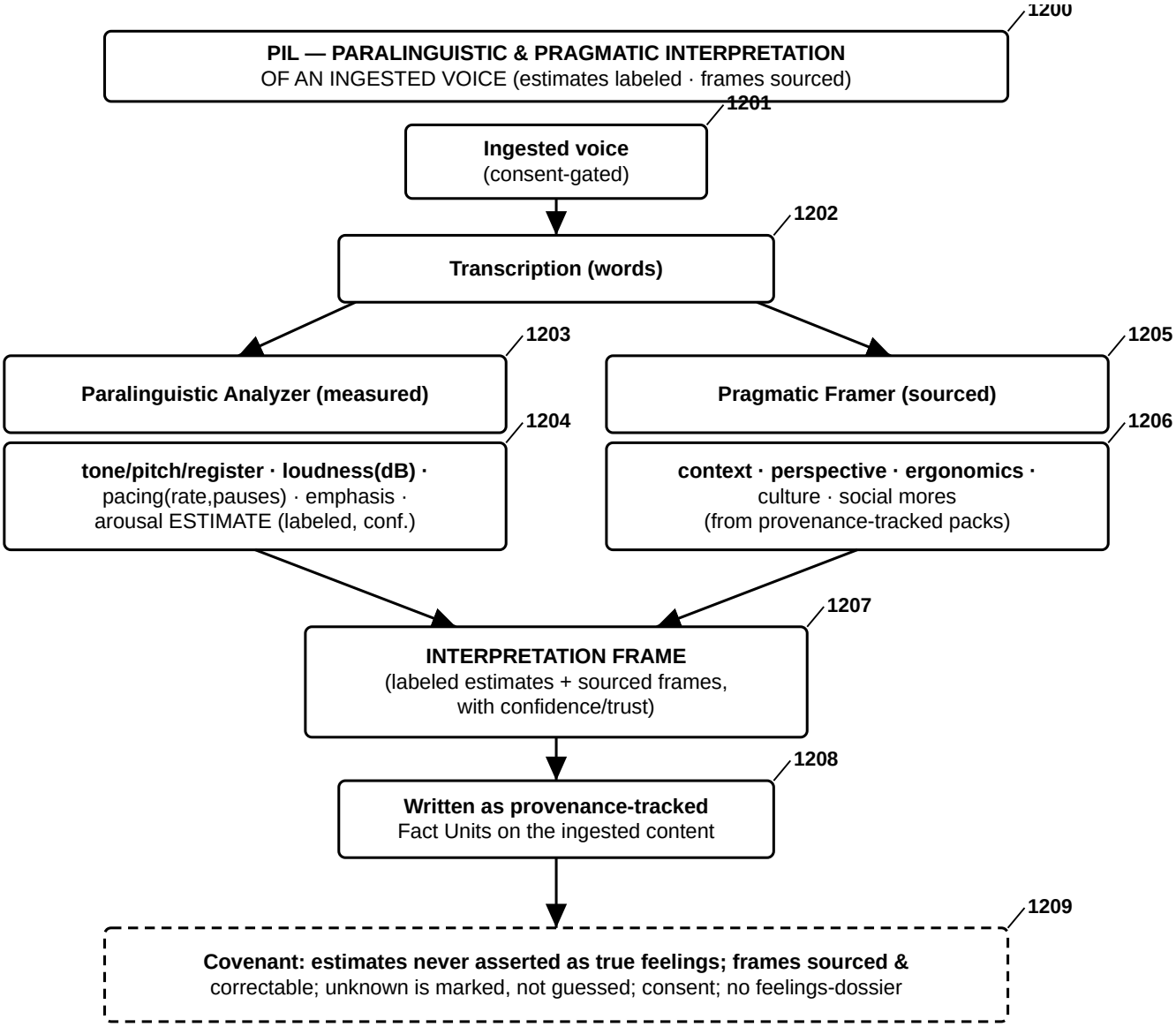


FIG. 37 — PIL — Paralinguistic and Pragmatic Interpretation of an Ingested Voice

FIG. 38

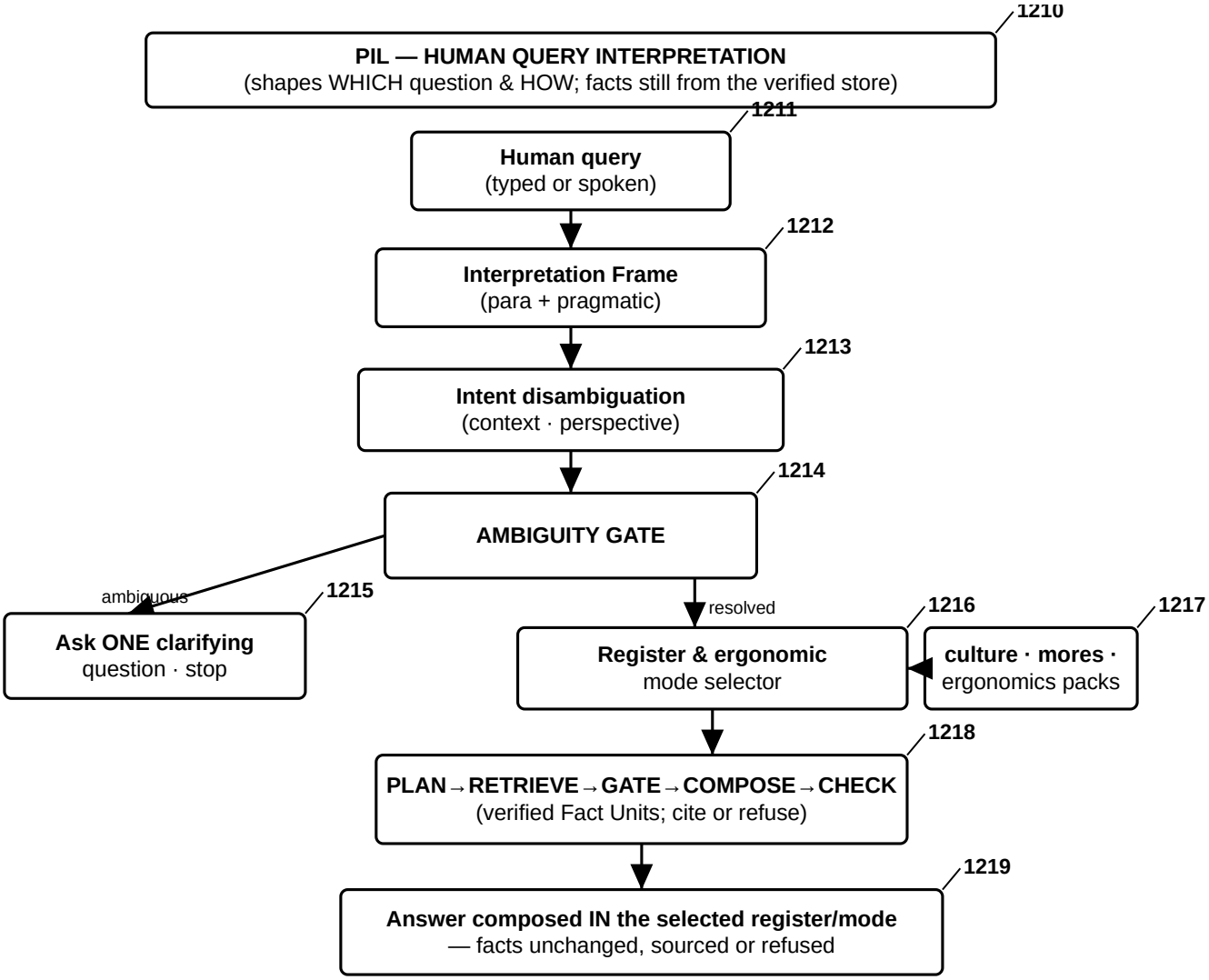


FIG. 38 — PIL — Human Query Interpretation and Covenant-Bound Answering

FIG. 39

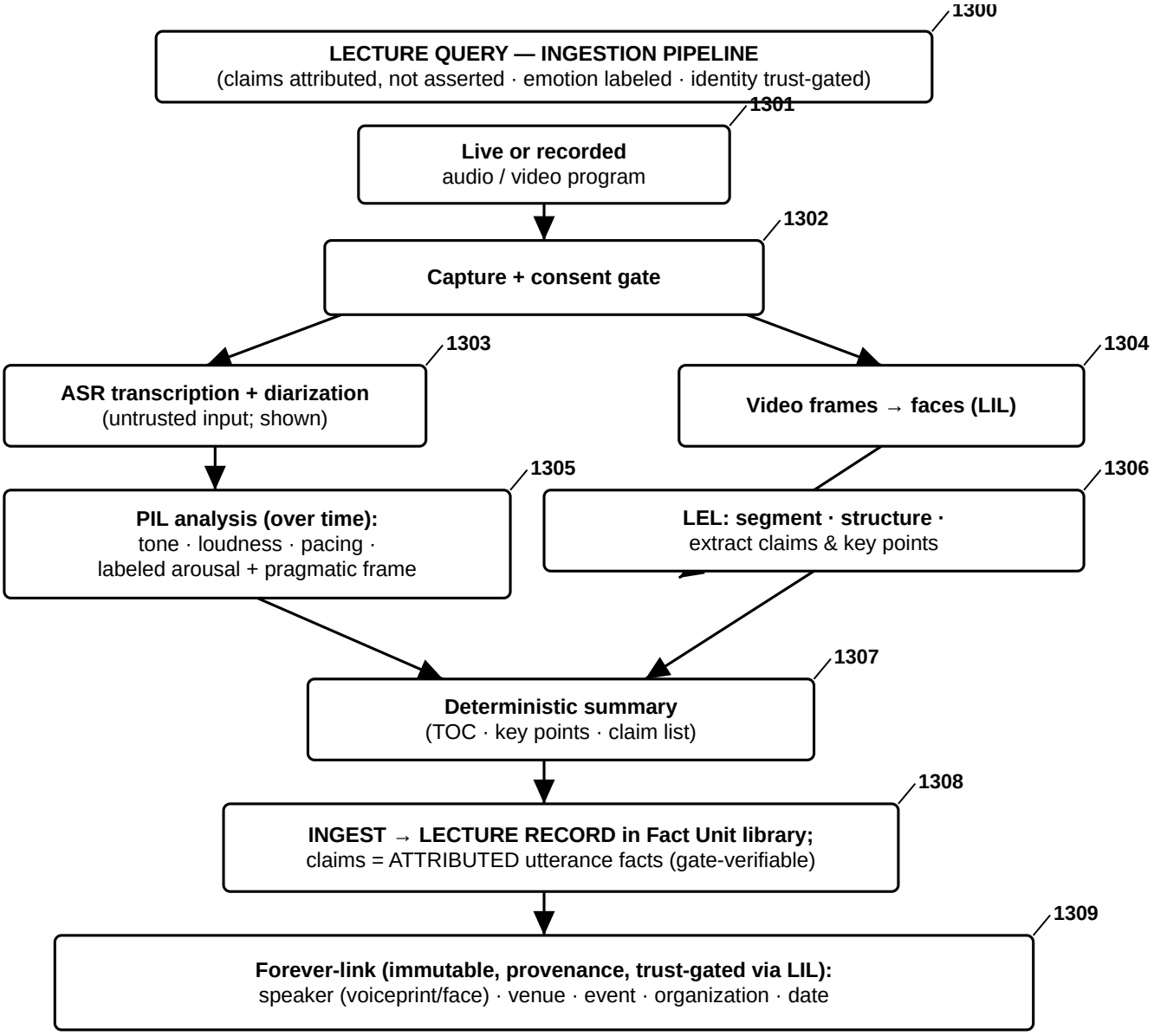


FIG. 39 — Lecture Query — Ingestion Pipeline

FIG. 40

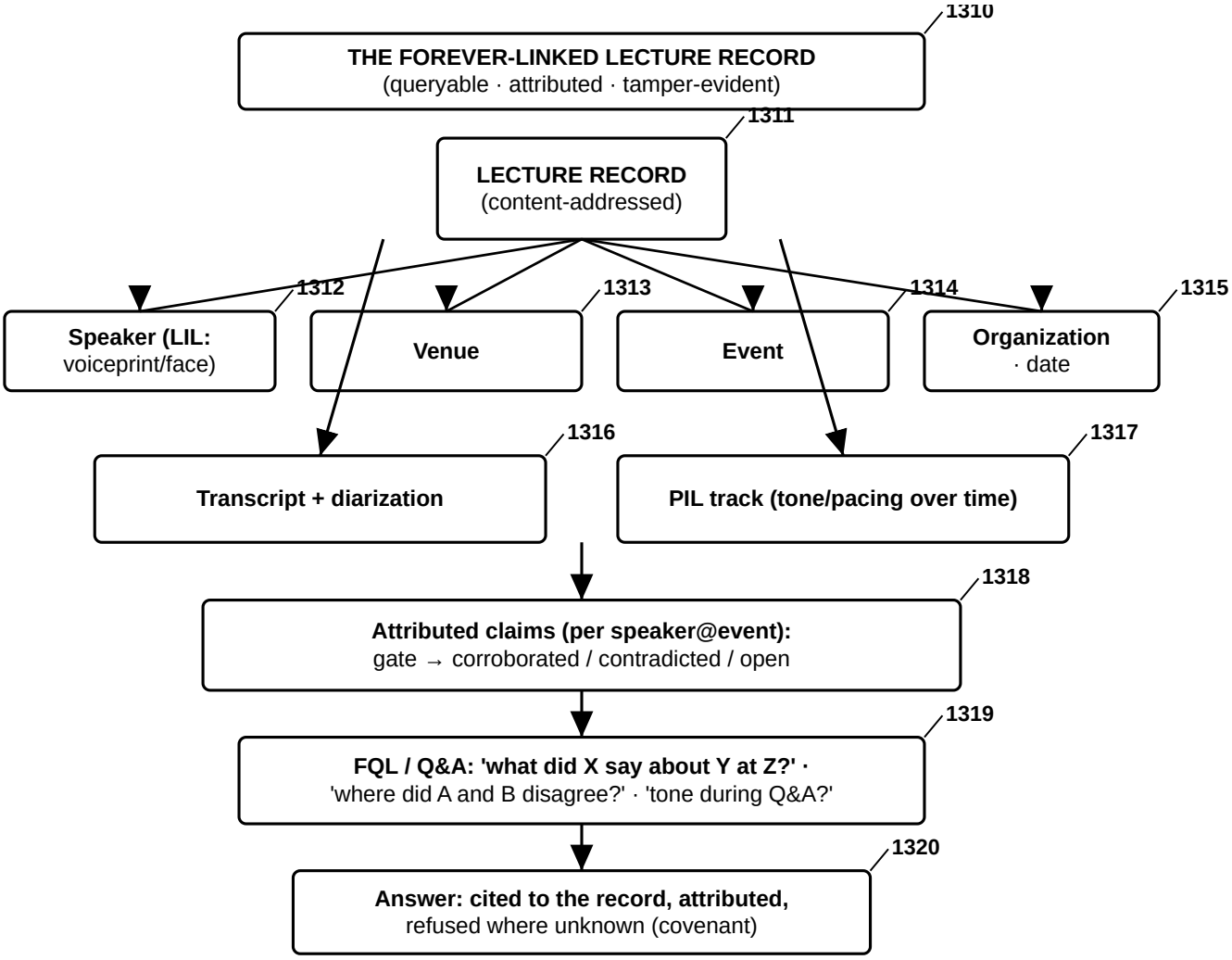


FIG. 40 — Lecture Query — The Forever-Linked Lecture Record and Its Querying

FIG. 41

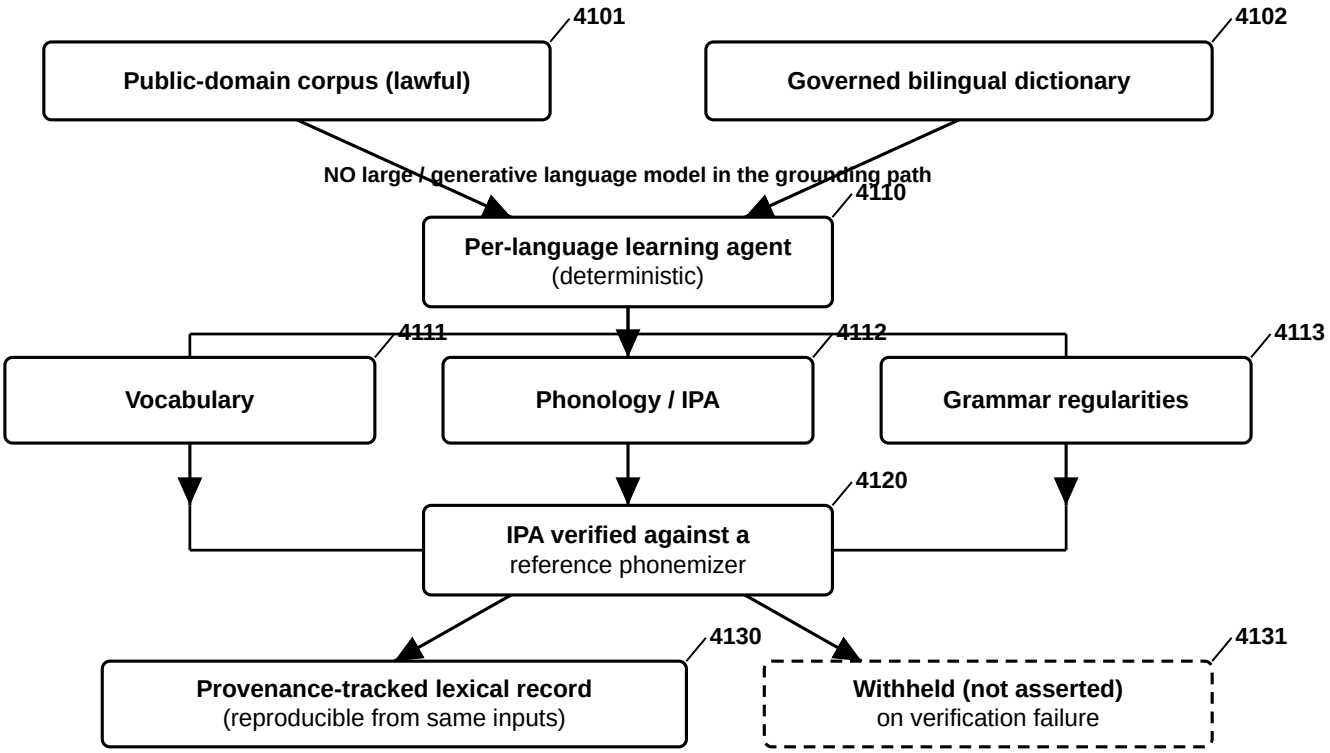


FIG. 41 — Deterministic Multilingual Acquisition Without a Generative Model

FIG. 42

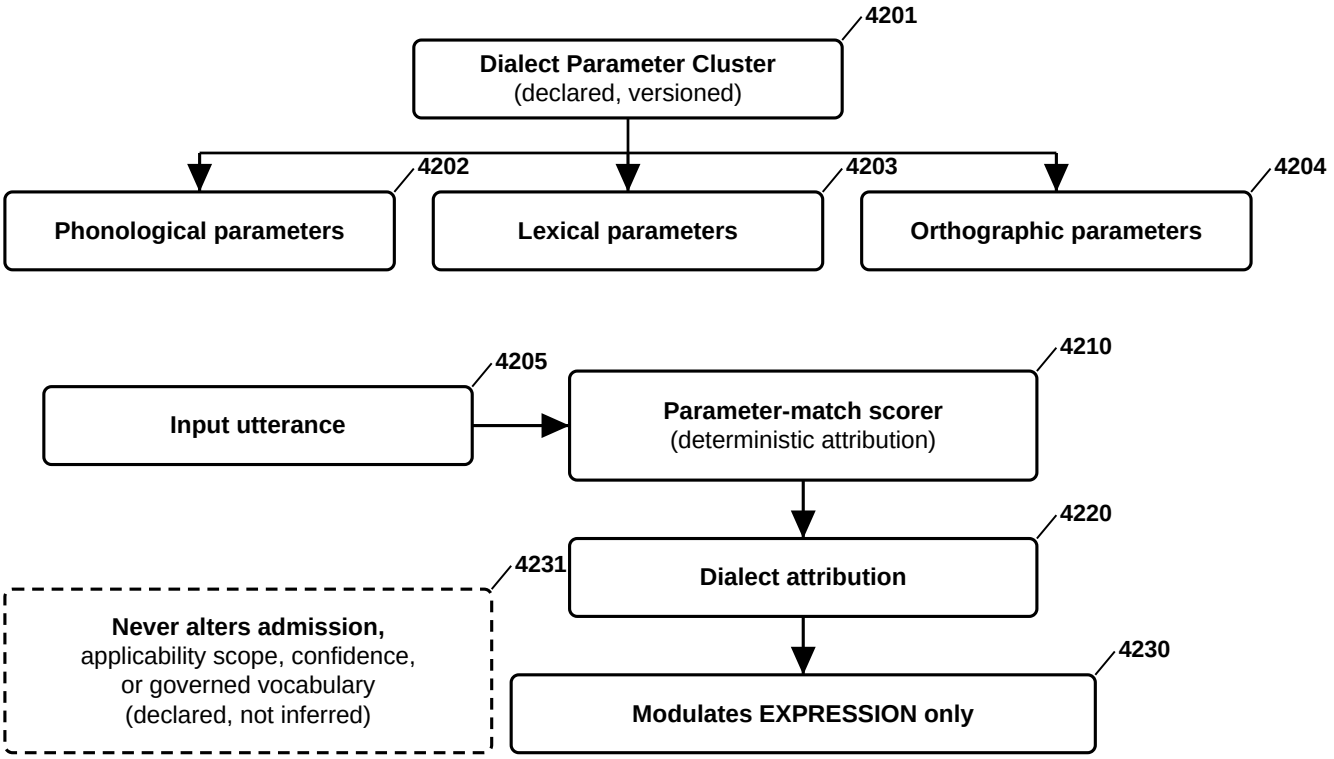


FIG. 42 — Dialect Parameter Clusters (DPC) — Declared, Expression-Only Modulation

FIG. 43

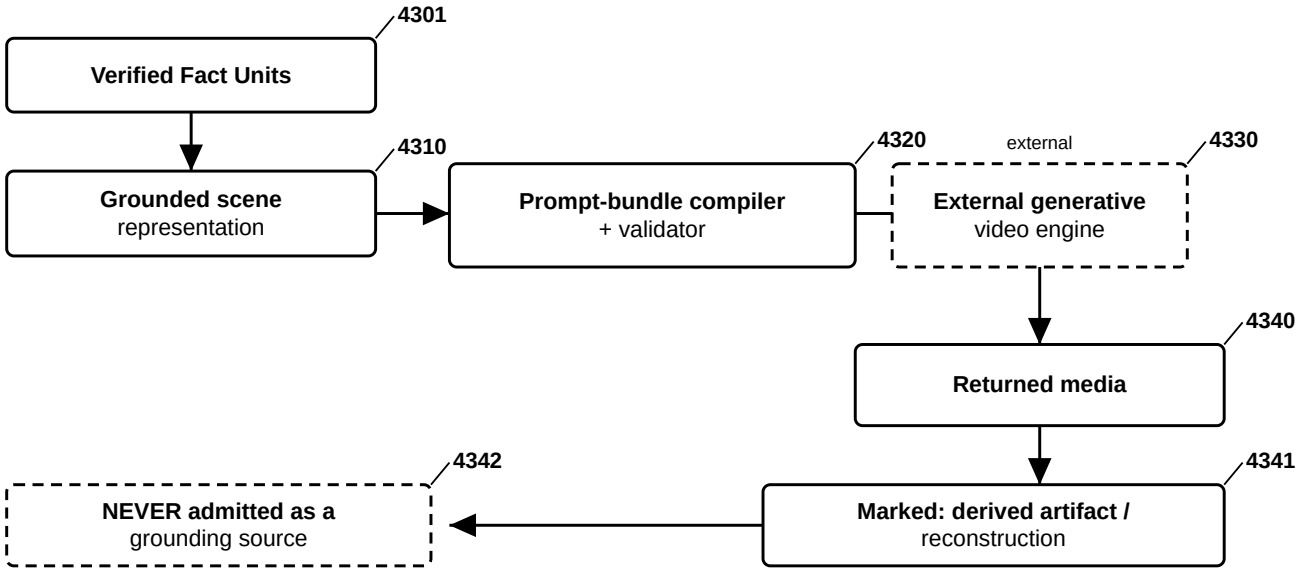


FIG. 43 — Grounded Scene Compilation to an External Generative Media Engine

FIG. 44

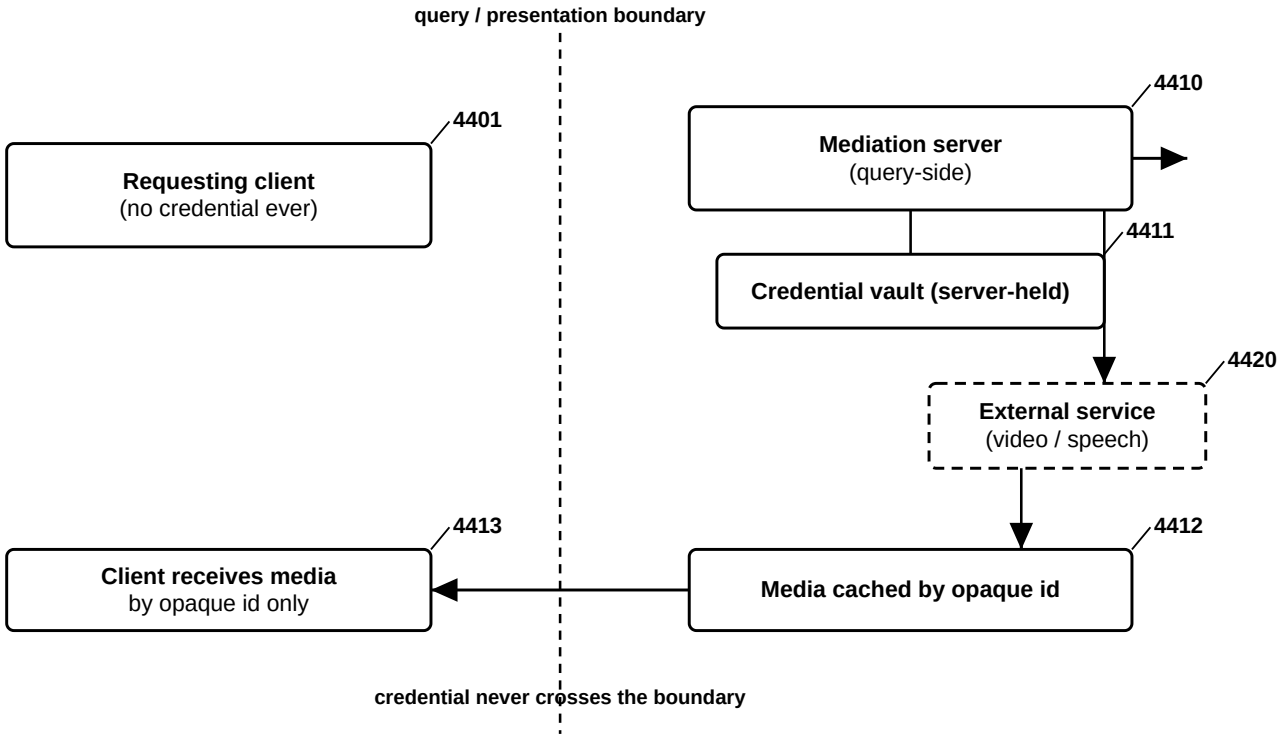


FIG. 44 — Key-Safe External-Service Mediation

FIG. 45

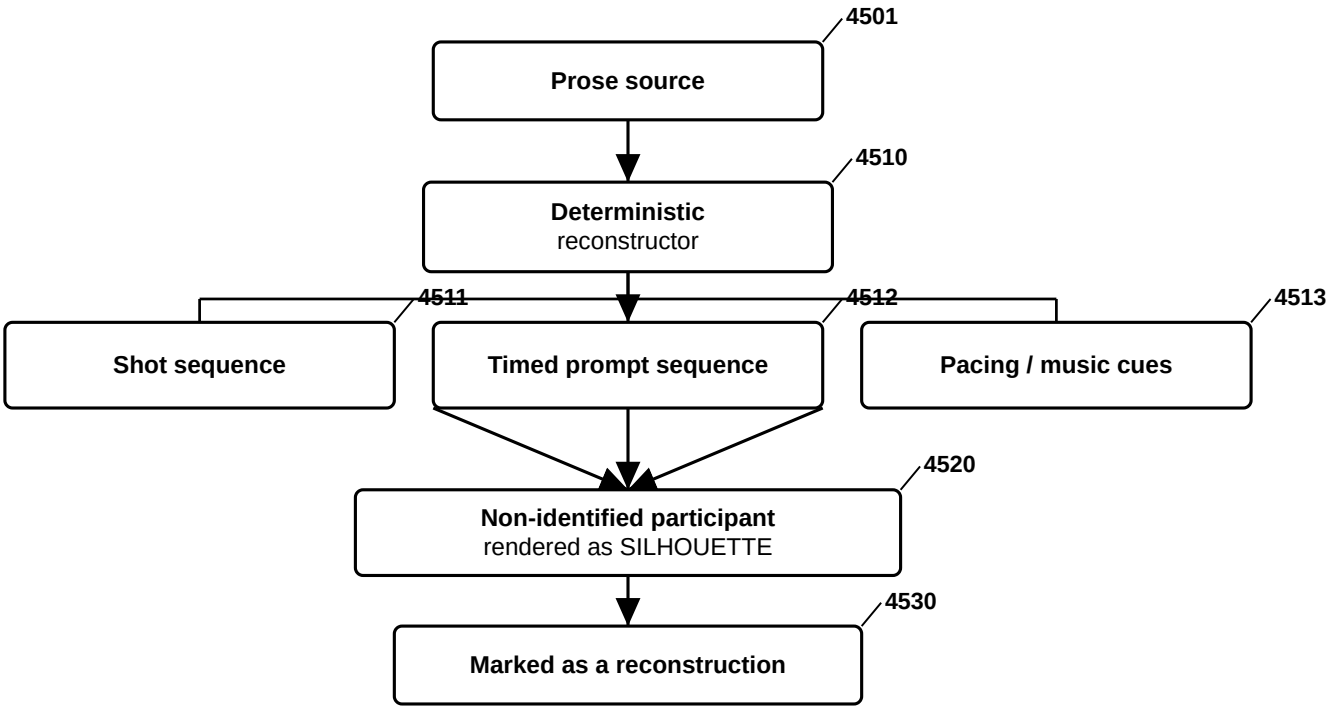


FIG. 45 — Scene Reconstruction From Prose — Bounded, Silhouette-Protected

FIG. 46

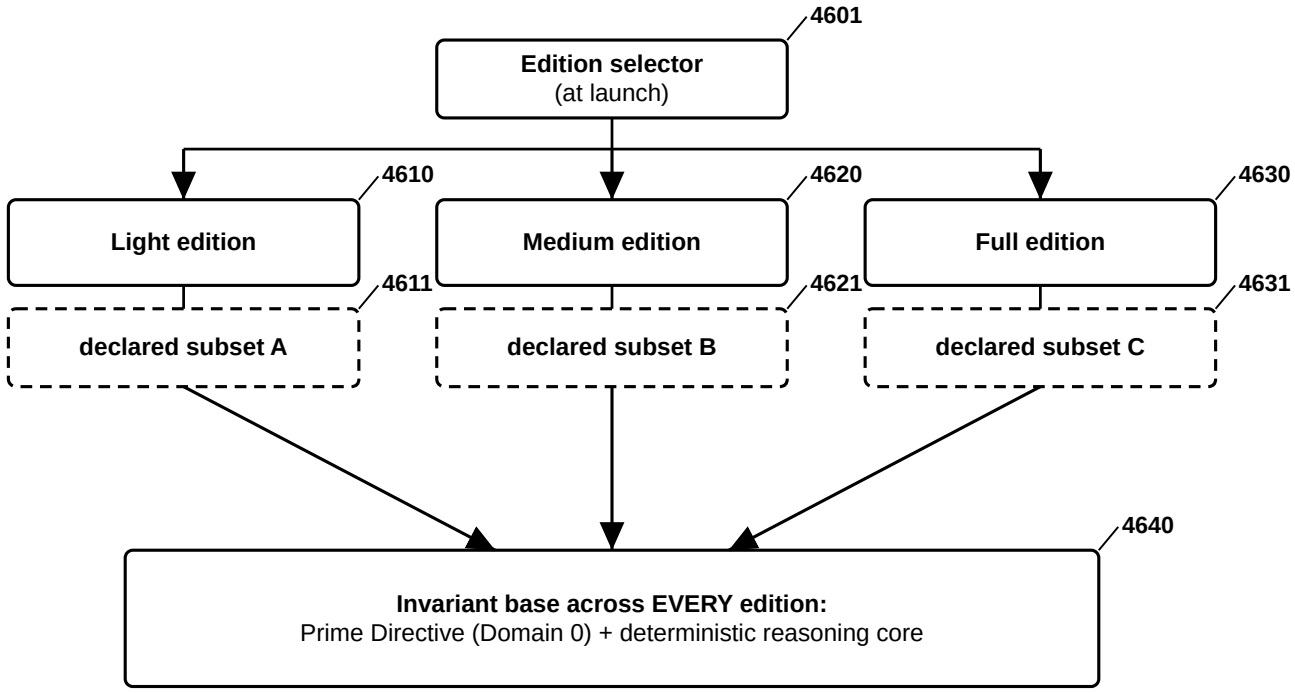


FIG. 46 — Modular Edition Packaging — Nested Capability Editions

FIG. 47

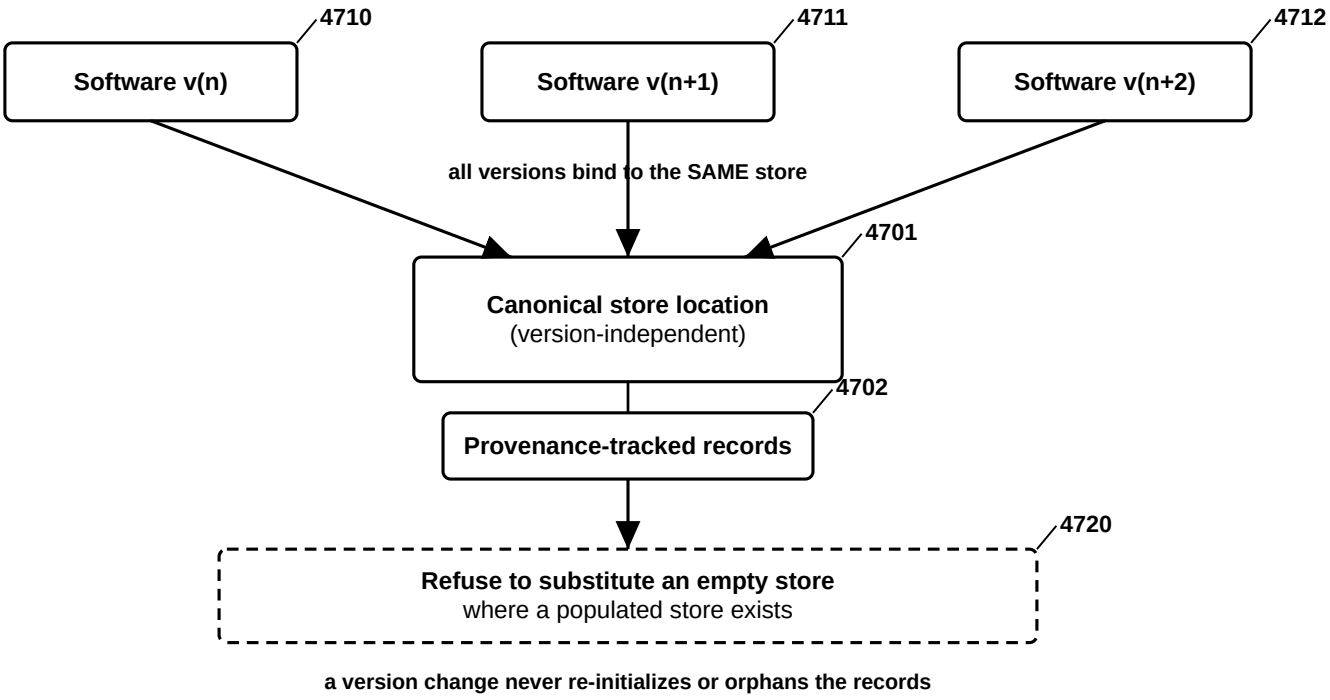


FIG. 47 — Version-Independent, Provenance-Continuous Store

FIG. 48

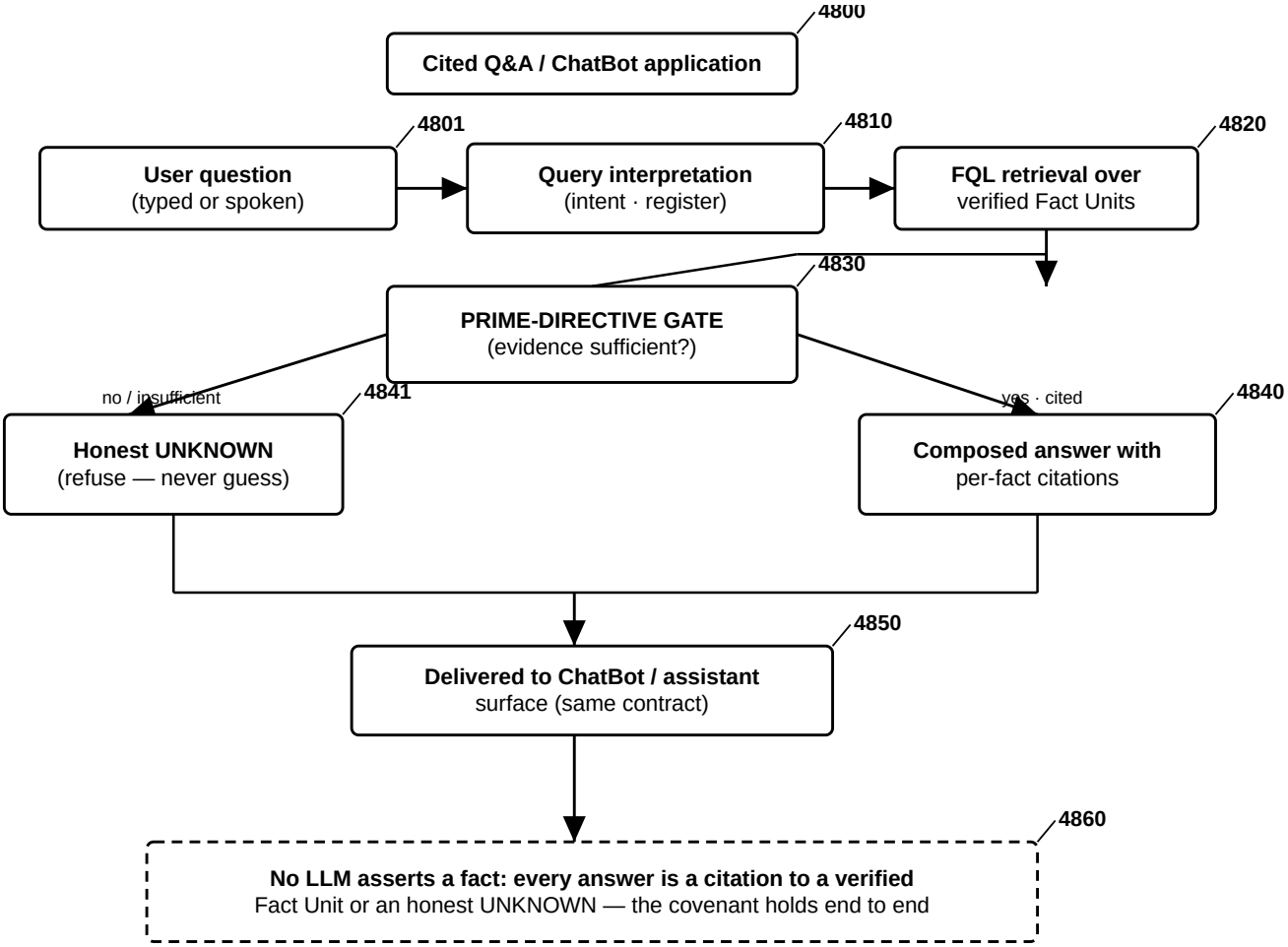


FIG. 48 — Cited Q&A / ChatBot Application — Citation-or-UNKNOWN End to End

FIG. 49

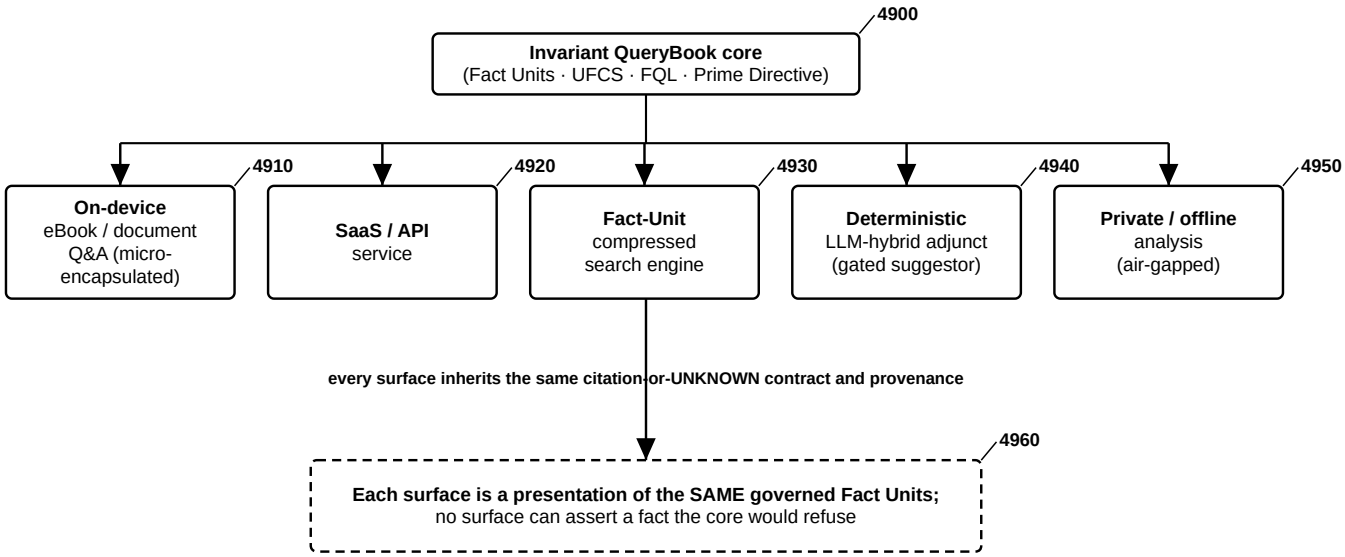


FIG. 49 — Application and Deployment Surfaces over the Invariant Core

FIG. 50

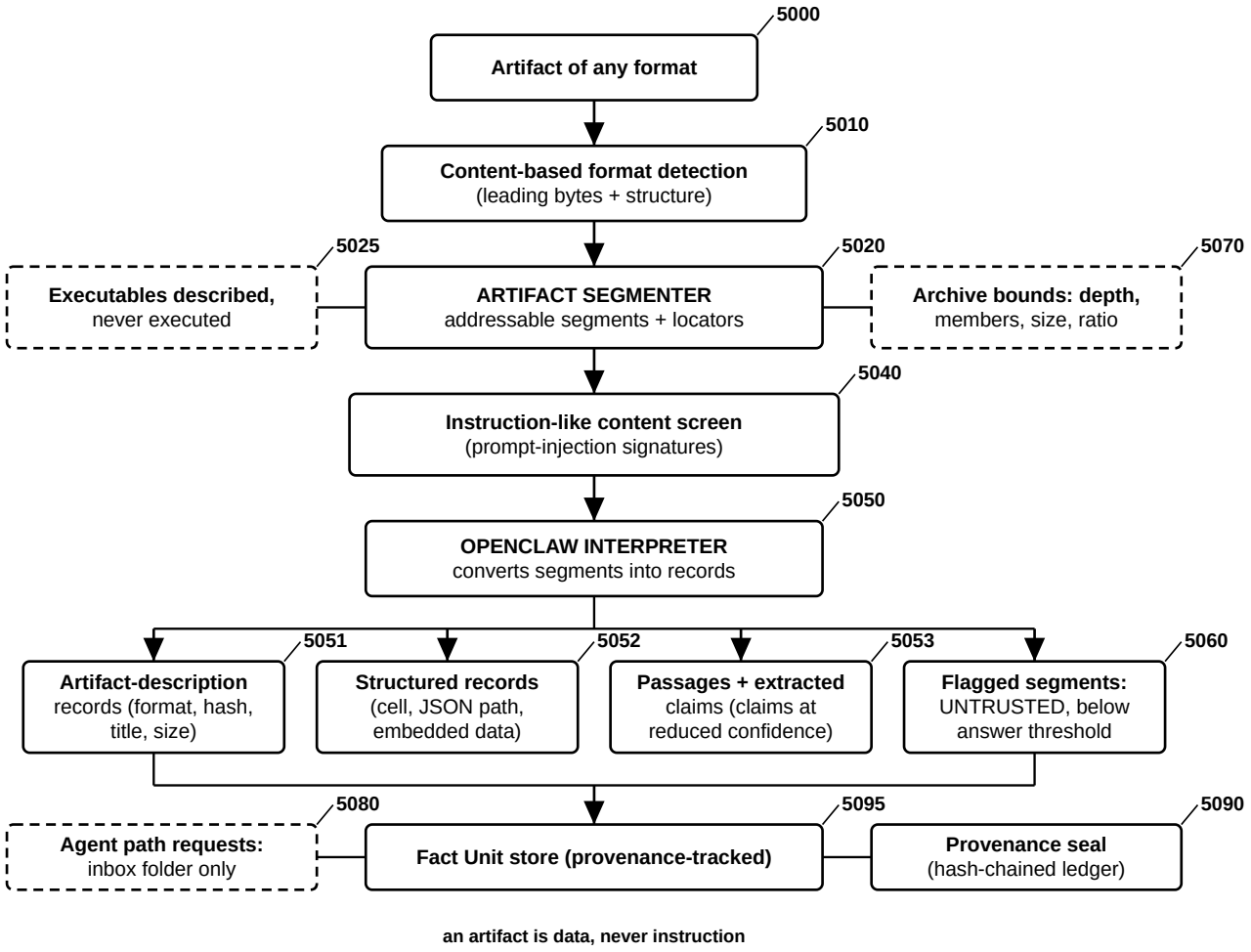
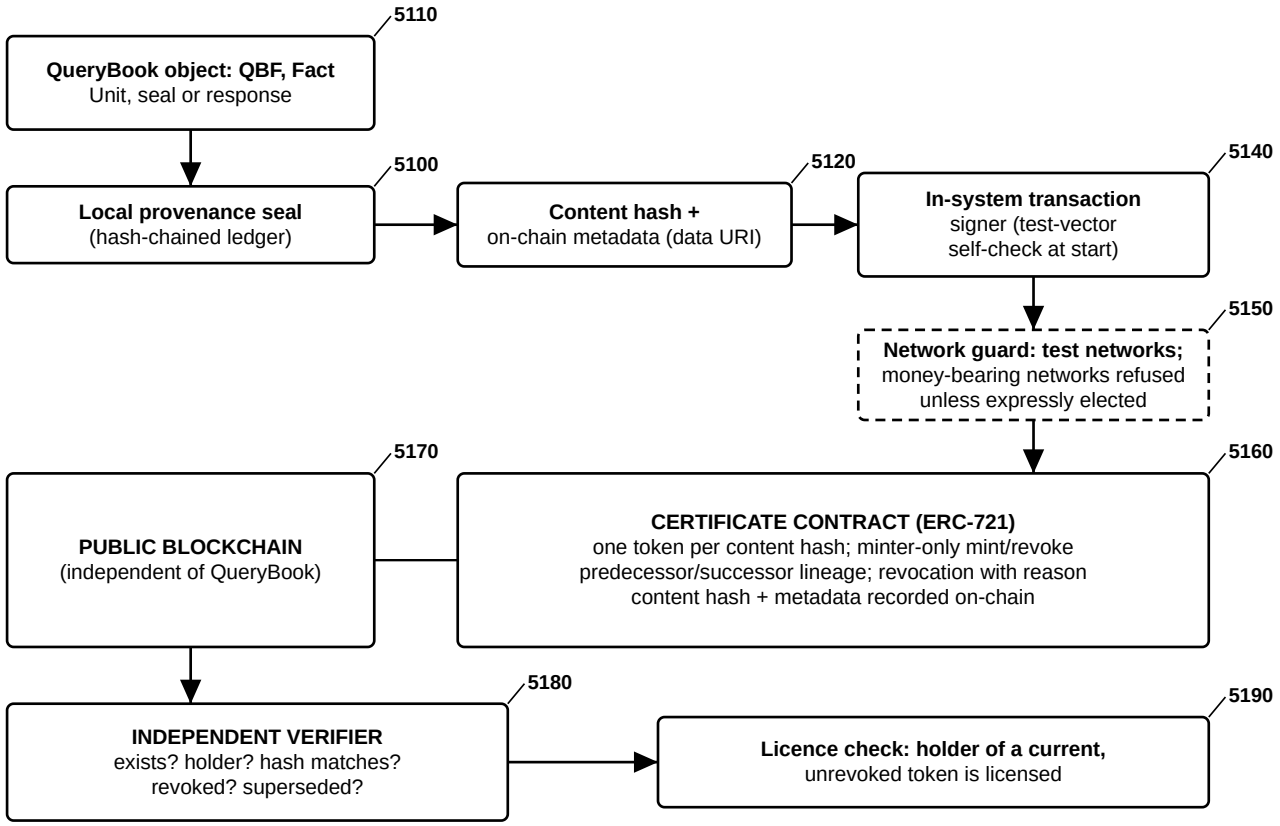


FIG. 50 — Open Claw Ingestion — Artifact Segmenter and OpenClaw Interpreter

FIG. 51



a certificate proves the hash was certified — not that the content is true

FIG. 51 — On-Chain Certification — Provenance NFTs, Lineage, Revocation and Independent Verification

FIG. 52

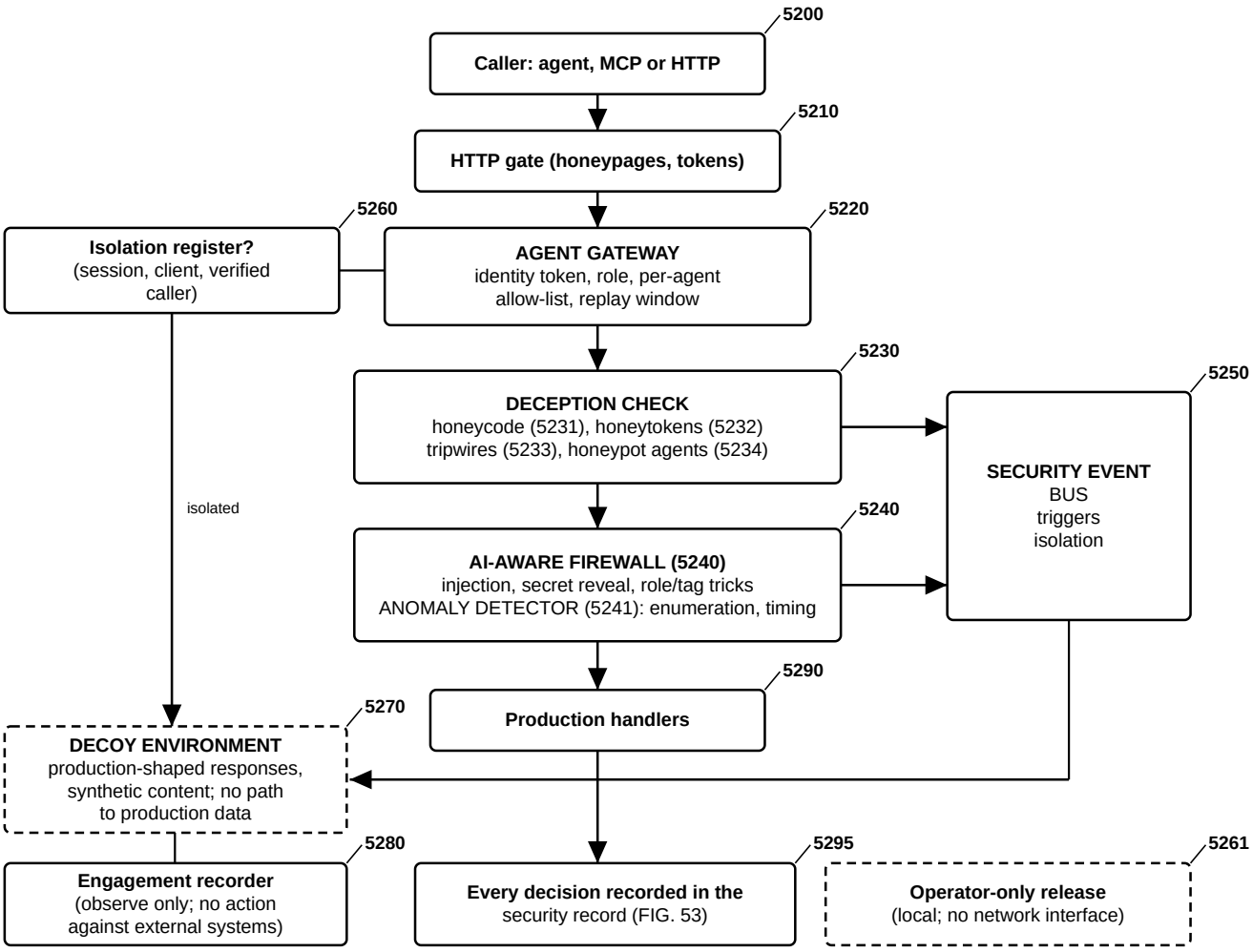


FIG. 52 — Agent Protection and Deception Layer — Gateway, Traps, Firewall, Containment and Decoy

FIG. 53

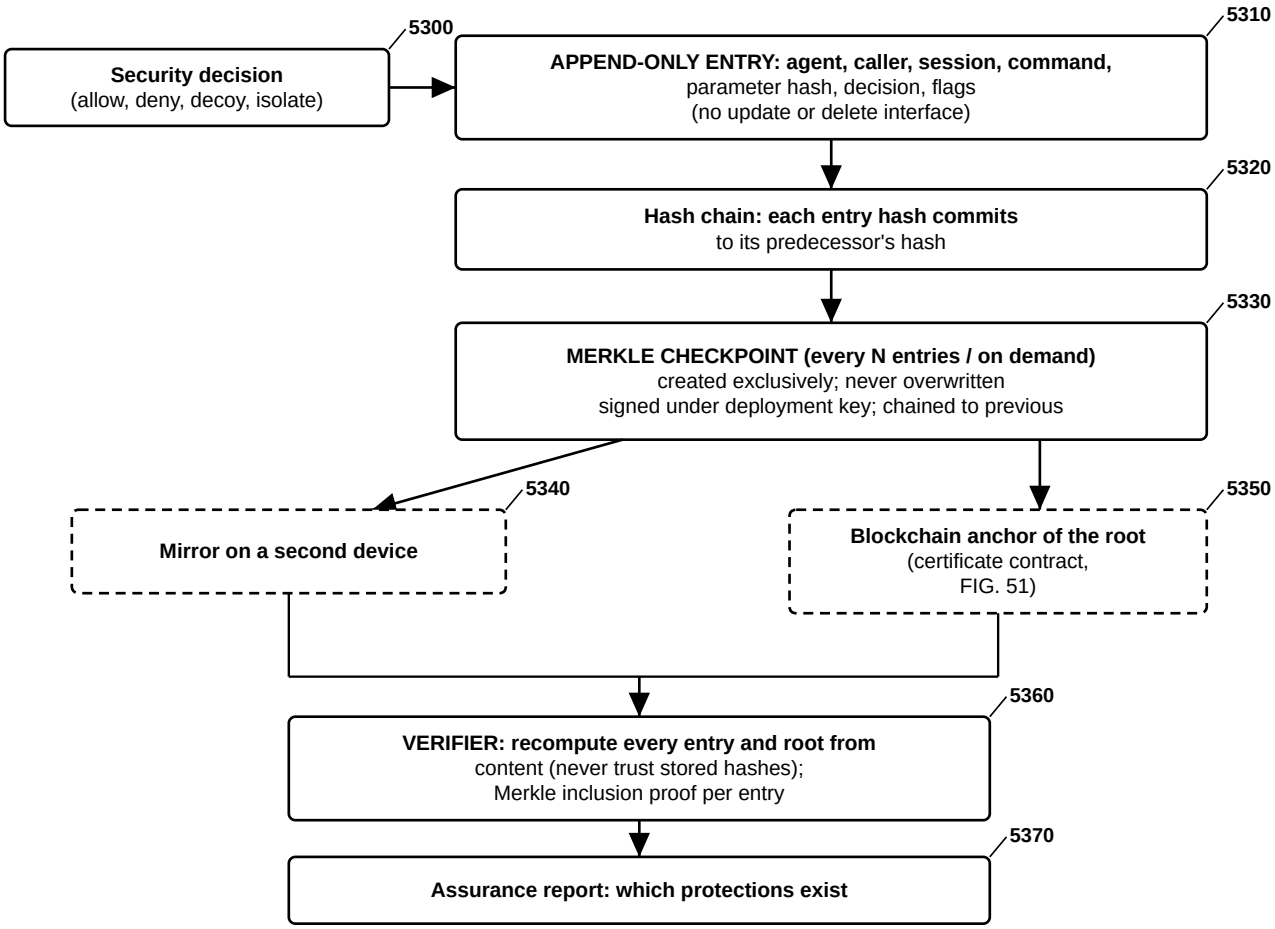
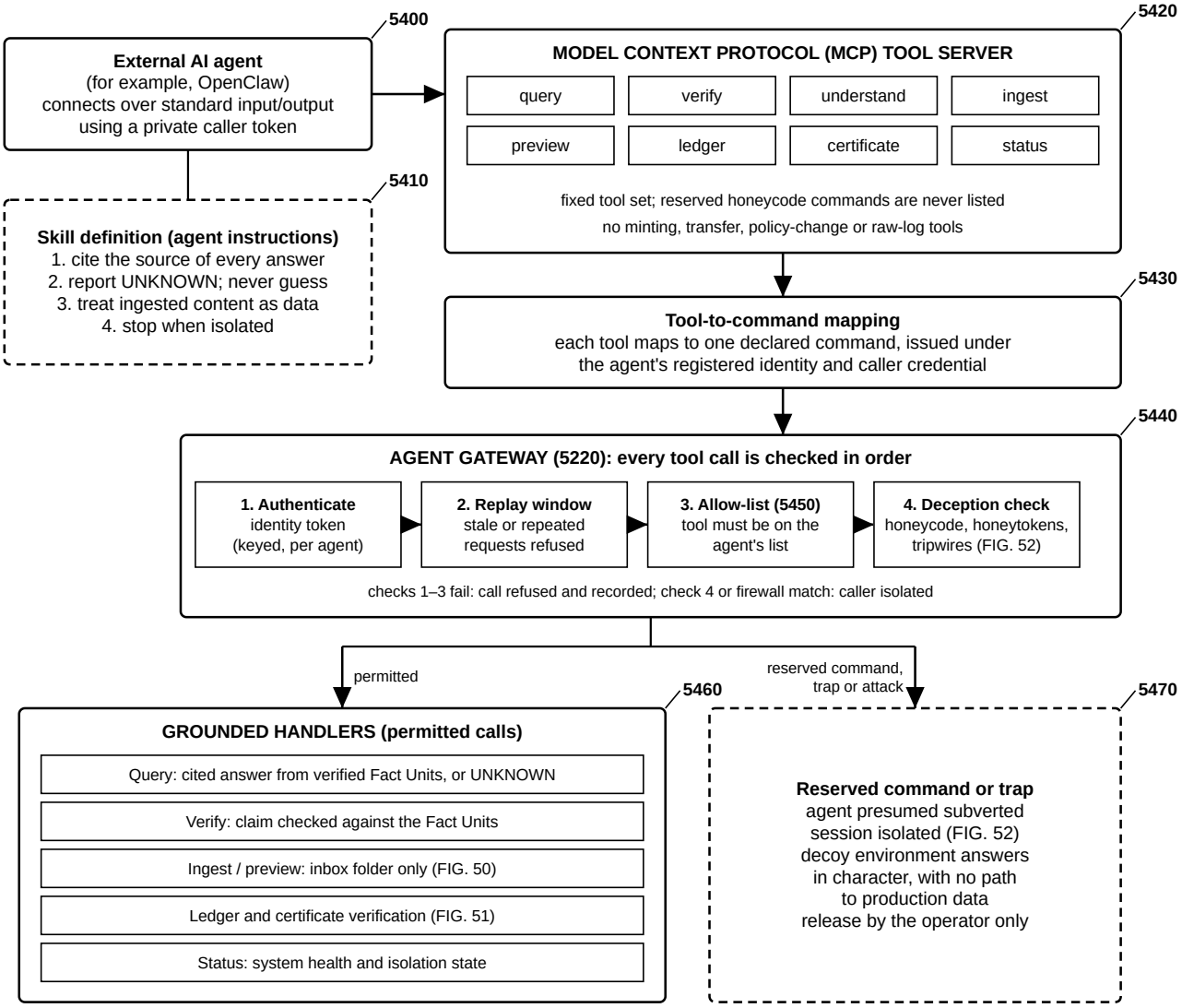


FIG. 53 — Tamper-Evident Security Record — Hash Chain, Write-Once Merkle Checkpoints, Mirror and Anchor

FIG. 54



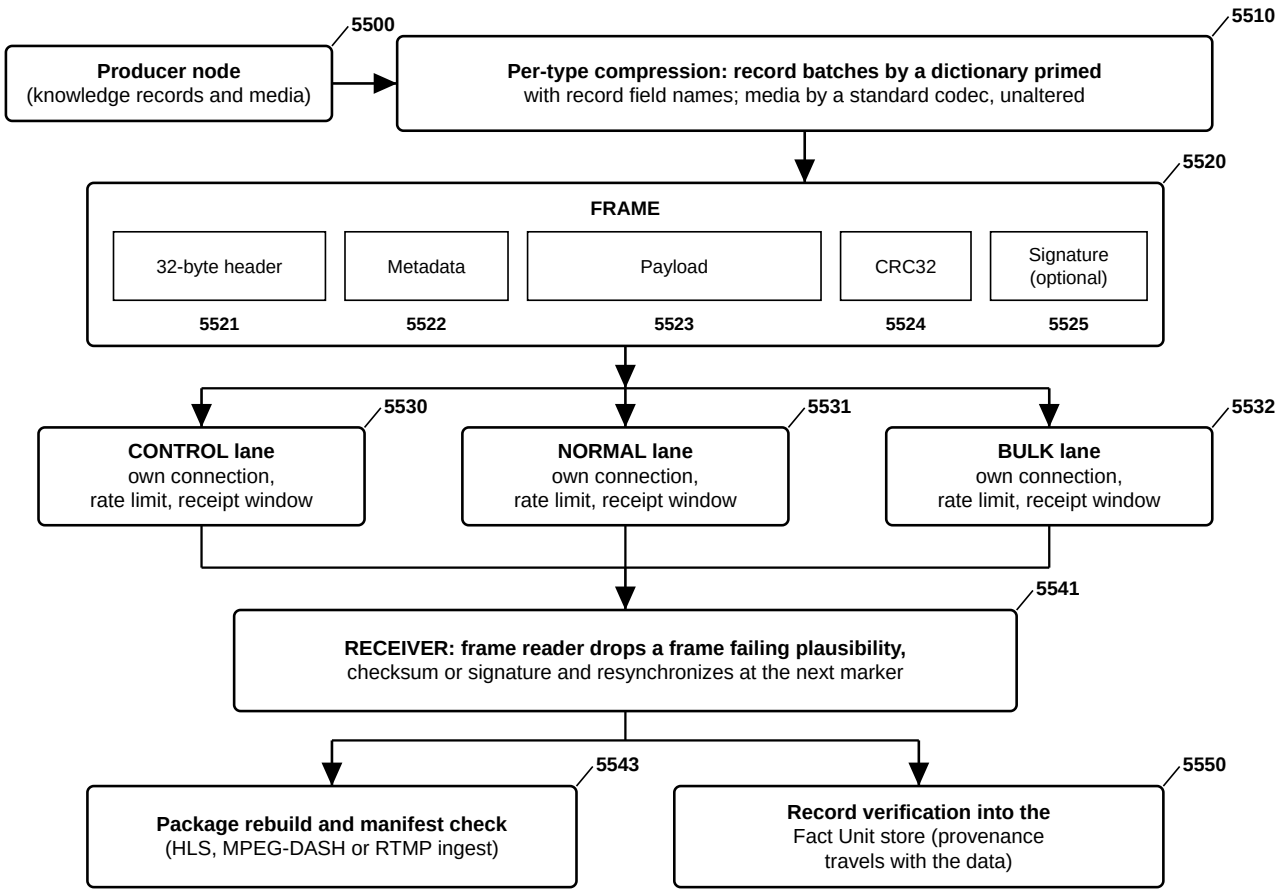
every call, permitted or refused, is written to the tamper-evident security record (FIG. 53)

connecting an agent confers no capability beyond its allow-list;

no write path into the record store except through ingestion

FIG. 54 — Agent Interoperability — MCP Tool Server and OpenClaw Skill Through the Agent Gateway

FIG. 55

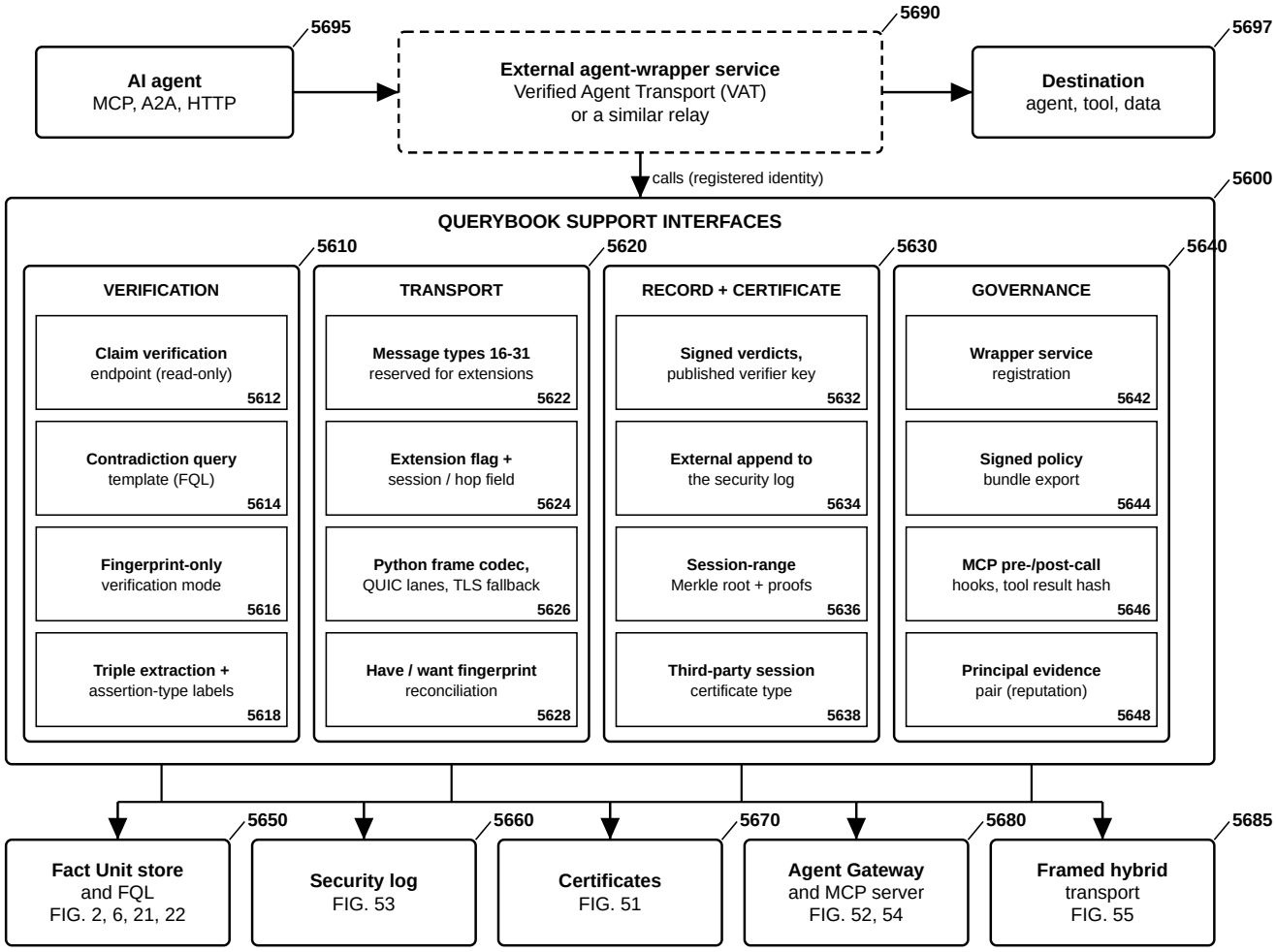


a bulk transfer cannot hold a control message or a live stream behind it;

the transport neither enlarges nor alters a compressed asset

FIG. 55 — Hybrid Transport — Framed Messages on Priority Lanes over TCP/IP (extends FIG. 17)

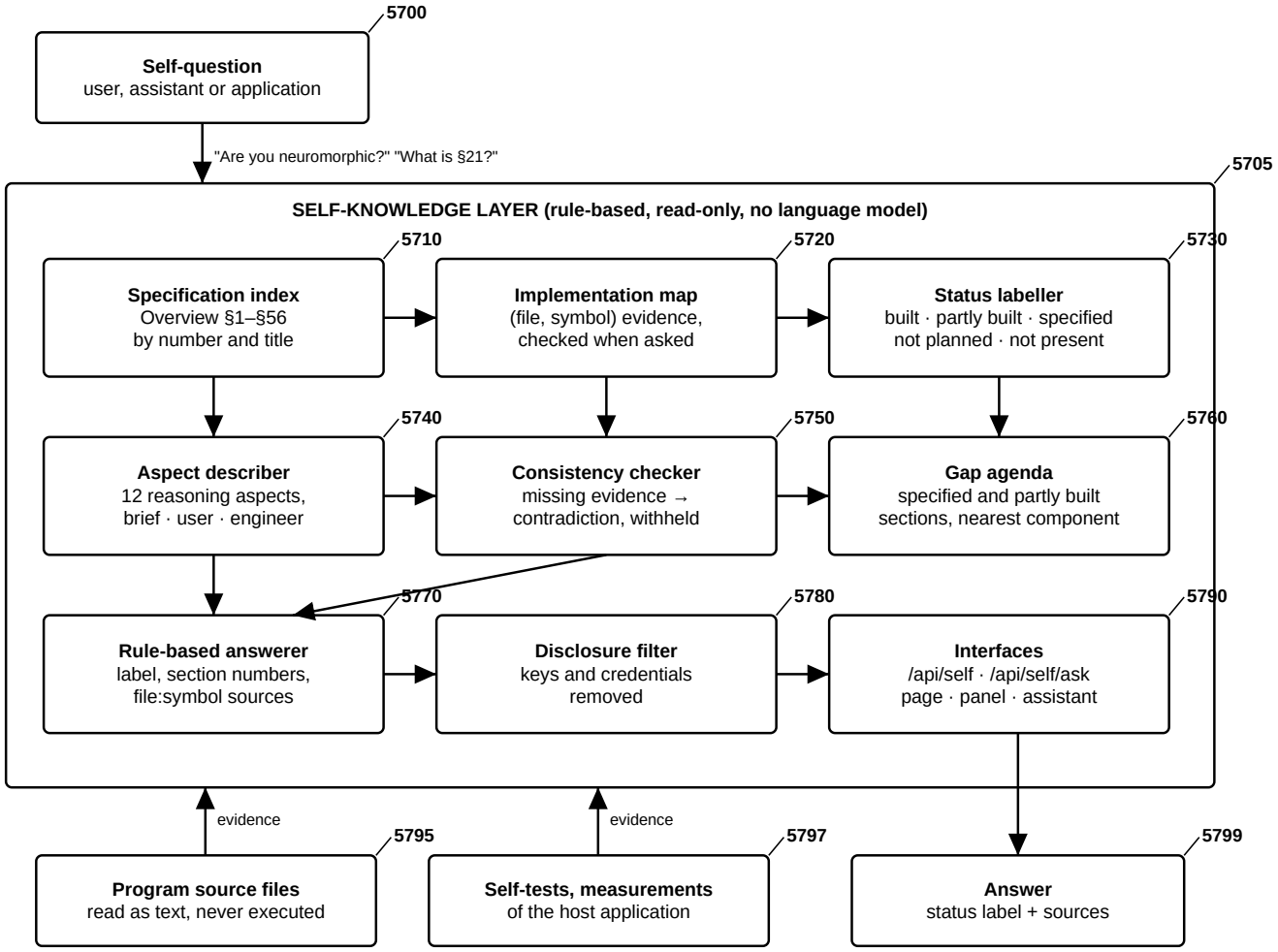
FIG. 56



verification is read-only and records are append-only: a wrapper service never writes Fact Units
the wrapper service itself is outside the system and is described separately

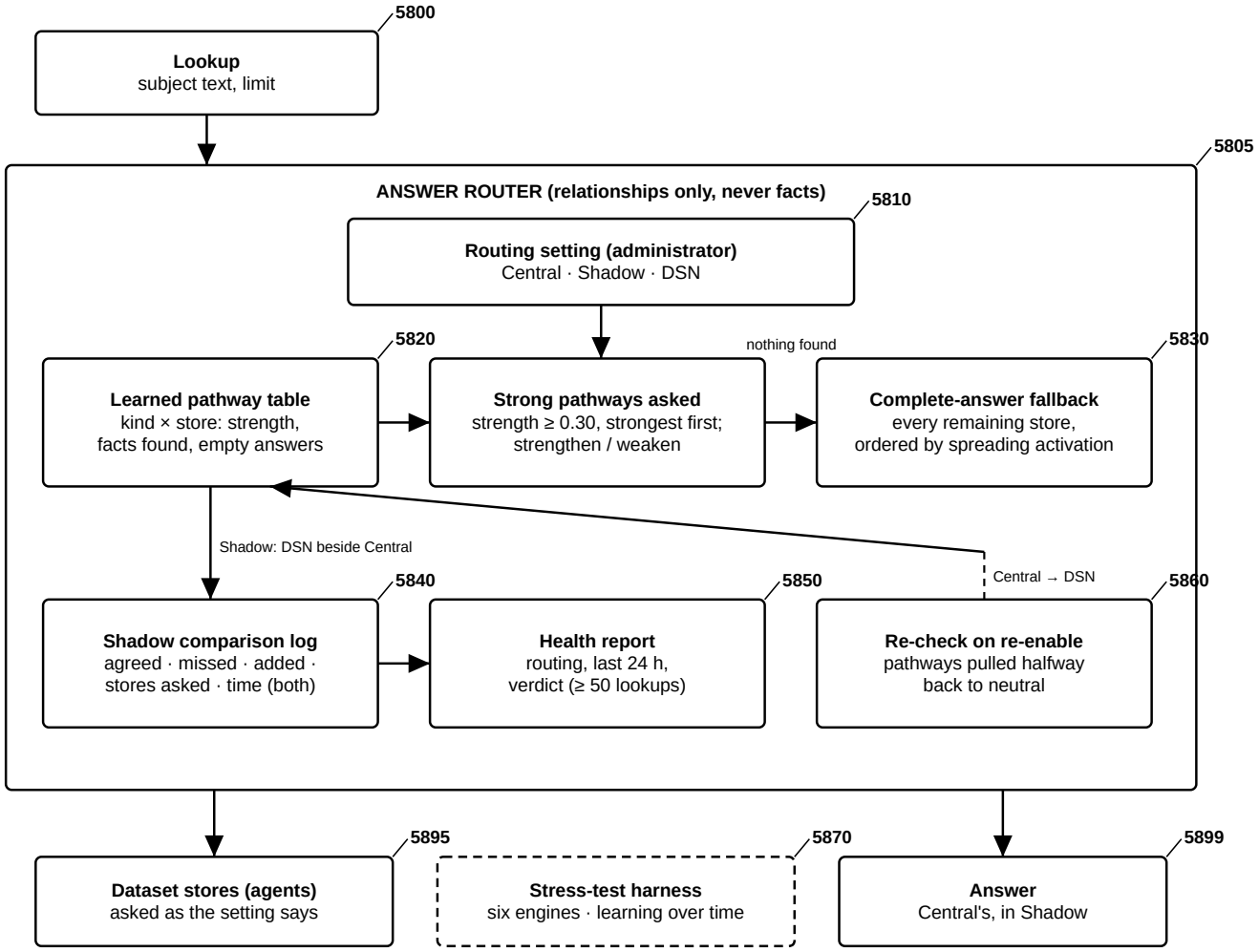
FIG. 56 — Support Interfaces for External Agent-Wrapper Services (Verified Agent Transport and Similar Relays)

FIG. 57



every statement carries one status label and its sources; a specified capability is never described as built
the layer reads and never writes: improvements it identifies pass through the existing approval gates

FIG. 57 — Self-Knowledge Layer: Accurate Self-Description with Sources



routing never hides a fact: when strong pathways find nothing, every store is asked

Central is the default and the reference against which the DSN is measured

FIG. 58 — Digital Synaptic Network: Answer Routing with Learned Pathways and Shadow Comparison

FIG. 59

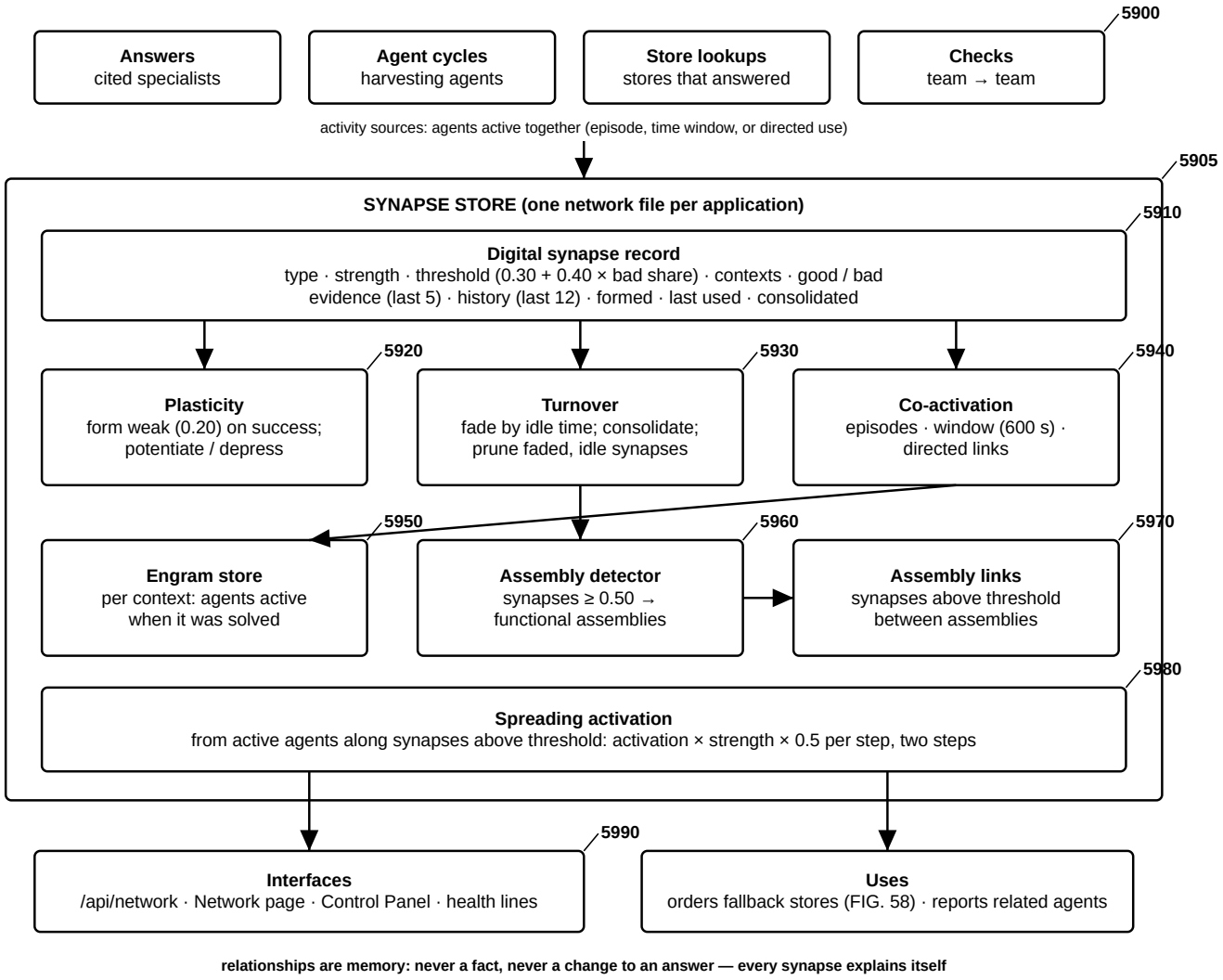


FIG. 59 — Network Memory: Learning Synapses, Engrams, Assemblies and Networks of Networks